

User Equivalent Range Error and Positioning Accuracy Analysis for ToA-Based Techniques Using PRS and SSB in 5G/6G NTN

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ABSTRACT The integration of Non-Terrestrial Network (NTN) within the Fifth Generation (5G) and Sixth Generation (6G) ecosystem offers new opportunities for global Positioning, Navigation, and Timing (PNT) services, particularly in environments where Global Navigation Satellite System (GNSS) signals are obstructed or degraded. This work investigates the positioning performance of Time of Arrival (ToA)-based techniques using both Positioning Reference Signal (PRS) and Synchronization Signal Block (SSB) signals in 5G NTN scenarios. A comprehensive User Equivalent Range Error (UERE) model is developed, incorporating satellite, receiver, and propagation-related error sources across FR1 and FR2 frequency bands. The influence of satellite geometry is assessed through a Geometrical Dilution of Precision (GDOP) analysis, while a minimum satellite visibility evaluation quantifies the likelihood of observing at least four satellites above various elevation masks for representative LEO constellations at 600 km and 1200 km altitudes. Monte Carlo simulations quantify the impact of UERE on positioning accuracy, revealing that the 95th-percentile horizontal errors range from approximately 8 m to 26 m, depending on signal type, frequency band, and orbital altitude. The results demonstrate that ToA-based positioning using PRS can surpass typical urban GNSS-level accuracy, highlighting its potential in GNSS-challenged environments. In comparison, SSB-based positioning remains a viable alternative, offering reduced complexity at the cost of moderate accuracy trade-offs. The study offers a unified framework for understanding positioning accuracy in 5G/6G NTN, bridging UERE modeling, satellite visibility, and signal design considerations, and supporting the ongoing development of reliable global PNT services.

INDEX TERMS 5G, 6G, GDOP, LEO, NTN, positioning accuracy, PRS, PNT, SSB, UERE.

I. INTRODUCTION

THE GLOBAL Navigation Satellite System (GNSS) has long been the cornerstone of Positioning, Navigation, and Timing (PNT) services, providing critical capabilities across a wide range of applications, from consumer-level navigation to high-precision positioning for military and industrial purposes. While traditional GNSS remains the primary positioning technology, the growing demand for enhanced accuracy, reliability, and availability, particularly in challenging environments such as urban canyons, deep indoor settings, and high-latitude regions, has driven

increasing interest in alternative and complementary PNT solutions. Among these, Low Earth Orbit (LEO)-based constellations have emerged as a promising technology due to their potential to improve performance in areas where GNSS signals are degraded or obstructed [1], [2], [3], [4].

Mobile communication networks have evolved with the advent of Fifth Generation (5G), which introduced significant advancements in connectivity, data rates, latency, and service continuity across diverse infrastructures [5], [6]. Standardized by 3rd Generation Partnership Project (3GPP), 5G initially focused on terrestrial networks, the 5G Terrestrial

Network (TN). Over time, 3GPP progressively integrated Non-Terrestrial Network (NTN) into its standardization efforts [7]. NTN includes a variety of spaceborne and airborne platforms such as Autonomous Aerial Vehicle (AAV), High-Altitude Platform Systems (HAPS), and satellites designed to extend connectivity to underserved or unserved regions. This integration was first formalized in 3GPP Release-17 [8], frozen in 2022, marking a key milestone in bridging non-terrestrial and terrestrial 5G ecosystems. Building on these developments, current research on Sixth Generation (6G) envisions even tighter integration of space-air-ground networks to enable ubiquitous, intelligent, and ultra-low-latency connectivity.

The potential of 5G/6G NTN as a positioning system has gained traction, particularly with the advent of LEO mega-constellations [9], which offer denser spatial coverage and lower latency than traditional GNSS Medium Earth Orbit (MEO) satellites. Within this framework, 3GPP standardization activities have explored the positioning capabilities of 5G NTN, aligning with the broader IMT-2030 vision for 6G, which promotes integrated connectivity with joint sensing and positioning capabilities [10]. Specifically, 3GPP Release-18 [11] introduced study item TR 38.882 [12] on network-verified User Equipment (UE) location, forming the foundation for positioning solutions in NTN scenarios. However, due to the prioritization of other topics, further enhancements in 5G NTN positioning were not included in 3GPP Release-19, leaving GNSS as the primary positioning method. Looking ahead, 3GPP Release 20, which is recognized as the first release initiating pre-studies for 6G, is expected to revisit and expand NTN-based PNT solutions in response to evolving use cases and industry-driven demands.

In parallel, recent 3GPP discussions have begun to address frequency band extensions relevant to NTN scenarios. Traditionally, Frequency Range 1 (FR1) and Frequency Range 2 (FR2) are defined as 410 MHz–7.125 GHz and 24.25 GHz–52.6 GHz, respectively, as specified in TS 38.104 [13]. However, Release-18 and beyond have explored potential extensions to support NTN configurations above 7.125 GHz and even beyond 71 GHz. This leads to a proposed mid-band designation, often referred to as Frequency Range 3 (FR3), covering approximately 7.125 GHz to 24 GHz, which is particularly suited for NTN due to improved propagation and reduced atmospheric losses compared to FR2 [14]. Furthermore, 3GPP has acknowledged interest in exploring even higher frequencies above 71 GHz, sometimes termed Frequency Range 4 (FR4), as part of long-term 6G research initiatives [15]. While neither FR3 nor FR4 are formally standardized at this stage, their consideration highlights the evolving nature of frequency range planning in support of NTN and beyond-5G networks.

3GPP has standardized several positioning techniques for 5G, including time-based methods such as Time Difference of Arrival (TDoA) and Time of Arrival (ToA), along with Angle of Arrival (AoA) and Enhanced Cell ID (E-CID), as

detailed in 3GPP TS 38.305 [16]. Each technique has specific applicability and constraints depending on network synchronization, antenna capabilities, and deployment scenarios. For example, AoA requires angle-resolving antenna arrays, while Observed Time Differential of Arrival (OTDoA) demands precise time synchronization between Next Generation Base Station (gNB). Time-based positioning methods are particularly suited to 5G NTN, where large propagation distances and clock synchronization challenges are inherent. This work focuses on ToA techniques, which provide a direct and scalable approach to range-based localization in NTN. The use of Positioning Reference Signal (PRS), benefiting from wider bandwidth and higher carrier frequencies, further improves the accuracy and robustness of ToA-based methods [17]. When integrated with existing GNSS systems, ToA-based PRS can enable hybrid positioning solutions that enhance coverage and reliability, especially in underserved regions.

In this study, building upon the current status of 3GPP advancements regarding 5G NTN positioning, we derive the User Equivalent Range Error (UERE) for 3GPP 5G/6G NTN scenarios. This derivation is intended to estimate the expected positioning accuracy when employing PRS in conjunction with ToA-based positioning techniques. In addition, we consider opportunistic positioning approaches that exploit the Synchronization Signal Block (SSB), which are particularly relevant when dedicated positioning signals are unavailable. The analysis provides insights into the achievable PNT performance and aims to guide future standardization and implementation efforts.

A. STATE-OF-THE-ART-REVIEW

Most existing research addressing positioning accuracy in LEO-based PNT systems primarily considers large-scale commercial satellite communication constellations, such as Starlink, Kuiper, and OneWeb. For instance, Ferre and Lohan [18] used error-based Geometrical Dilution of Precision (GDOP) to assess positioning accuracy when comparing different satellite constellation architectures. Their analyses assumed simplified error models, including atmospheric residual errors, multipath errors, and receiver-related inaccuracies. Under these assumptions, the authors reported 1-sigma positioning errors ranging between 1.5 m and 3.2 m, highlighting the strong potential of large commercial constellations for accurate PNT services.

In the context of emerging 5G NTN systems, despite being in an early research stage, several recent studies have begun quantifying expected positioning performance. Typically, these studies employ system-level simulations, incorporating satellite constellation layouts and simplified channel propagation models. For example, Dureppagari et al. [9] investigated the positioning accuracy achievable using multi-satellite and single-satellite LEO-based NTN deployments. Their results suggest that, under favorable propagation conditions and with precise synchronization, sub-decimeter positioning accuracy is attainable in multi-satellite NTN scenarios. Conversely, accuracy degraded substantially in

TABLE 1. Summary of related works on UERE modeling, ToA-based positioning, and 5G/NTN positioning.

Reference	System / Signal	Methodology	Reported Perf.	Key Limitations
Ferré <i>et al.</i> [18]	LEO comm. constellations	GDOP-based accuracy with simplified error models (atmospheric residuals, multipath, receiver errors)	1.5–3.2 m (1σ)	No complete UERE; simplified assumptions
Dureppagari <i>et al.</i> [9]	5G NTN (multi-/single-sat)	System-level simulations with simplified channels and limited satellite counts	<10 m (multi-sat); sub-km (single-sat)	No detailed error model; small/idealized geometries
Emara <i>et al.</i> [19]	5G NTN with PRS	Simulated positioning under various constellations; emphasis on dedicated PRS	2–4 m (2σ)	No iono/tropo/multipath terms; optimistic assumptions
Lapin <i>et al.</i> [20]	Live 5G NR-NTN PRS (Ka, 100 MHz)	Experimental ToA ranging from on-orbit signals	<0.1 m (ranging, high SNR)	No full positioning; other error sources not modeled
Brambilla <i>et al.</i> [21]	Terrestrial 5G SSB	ToA-based localization using SSB in terrestrial deployments	Sub-decimeter (terrestrial)	No NTN validation; terrestrial-only channels
Camajori <i>et al.</i> [22]	Terrestrial 5G SSB	Feasibility analysis with current cellular deployments	Sub-decimeter (terrestrial)	No NTN dynamics; simplified assumptions
Kassas <i>et al.</i> [23]–[27]	Terrestrial 5G (opportunistic)	Always-on / on-demand ToA estimation; receiver design; vehicular studies	Sub-decimeter (terrestrial)	No LEO/NTN effects; not PRS/SSB in NTN

single-satellite scenarios, resulting in sub-kilometer-level performance. However, the authors did not detail the underlying error models employed in their analysis. Moreover, their investigation was limited to simplified scenarios with a maximum of seven satellites arranged in optimized geometries, rather than realistic full-scale constellation configurations.

Further contributions have explored the influence of satellite constellation configurations and dedicated positioning signals on achievable accuracy. Emara [19] assessed positioning performance considering various satellite deployment scenarios and emphasized the critical role played by dedicated PRS signals and constellation geometries. Their simulations indicated promising 2-sigma positioning accuracies between 2 m and 4 m. However, important practical sources of error, such as residual ionospheric errors and multipath propagation effects, were not included in their simulation models. Consequently, the reported accuracies may be overly optimistic and could deviate from real-world operational performance.

Significant experimental progress was achieved by Lapin *et al.* [20], who presented, for the first time, code-phase range measurements derived from real-world live signals of a 5G NR-NTN satellite system operating in the Ka-band. Their experiments involved a Telesat LEO satellite transmitting dedicated PRS with a 100 MHz bandwidth. The authors focused exclusively on the ToA measurement error, achieving an impressive sub-decimeter ranging accuracy for positive Signal-to-Noise Ratio (SNR). However, the scope of their work was limited, omitting several critical factors that typically influence the ranging measurement performance, such as ionospheric delays and constellation geometry when computing the position. Consequently, generalizing these impressive ranging accuracies to realistic positioning scenarios remains challenging without comprehensive modeling and analysis of additional error sources.

Beyond the use of dedicated PRS, recent research has increasingly investigated opportunistic positioning leveraging the SSB broadcast in terrestrial 5G networks. Several studies have demonstrated the feasibility of using SSB signals for positioning, achieving sub-decimeter accuracies under favorable conditions. For example, Brambilla *et al.* [21]

analyzed SSB-based time-based localization in terrestrial environments, while Camajori Tedeschini *et al.* [22] evaluated the feasibility of SSB-based positioning using current cellular network deployment. A significant body of work in this domain has been led by Prof. Kassas and his team, who have investigated opportunistic navigation and ToA estimation using always-on 5G downlink signals. Notably, their contributions include the design of receivers and ToA estimation algorithms for opportunistic localization with 5G signals [23], the exploitation of always-on and on-demand 5G signals for vehicular positioning [24], and the development of systems and methods for opportunistic ToA estimation [25]. Additional studies have explored the use of on-demand 5G downlink signals for opportunistic navigation [26] and cognitive navigation strategies in private networks using 5G and beyond signals [27].

However, it is important to note that the majority of existing studies focus exclusively on terrestrial 5G scenarios, with no direct experimental investigations of SSB-based positioning in NTN environments to date. The significantly different propagation characteristics, larger Doppler shifts, and sparser SSB transmission patterns in NTN, particularly with LEO satellite constellations, pose new challenges. As such, SSB-based positioning in NTN remains largely an open research area, necessitating further studies to adapt, optimize, and validate these techniques for satellite-based systems.

A summary of the reviewed contributions, including system characteristics, methodology, reported performance, and limitations, is provided in Table 1.

Overall, while these preliminary results highlight the significant potential and feasibility of accurate positioning using 5G NTN signals, both through dedicated PRS and opportunistic SSB-based techniques, it remains clear that comprehensive modeling and empirical validation of complete error sources, including synchronization accuracy, atmospheric and multipath errors, and realistic constellation configurations, are essential for reliably estimating achievable accuracy in real-world deployments.

B. PAPER CONTRIBUTIONS

To the best of the authors' knowledge, this article is the first to analyze the UERE budget in the context of 5G NTN

positioning considering both PRS and SSB signals. In this article, we provide a detailed analysis of the UERE budget and its impact on positioning accuracy in realistic 5G NTN scenarios. The primary contributions can be summarized as follows:

- 1) *UERE Modeling and Derivation*: A complete breakdown of major error sources is presented, including satellite clock and ephemeris inaccuracies, ToA measurement uncertainty, multipath effects, ionospheric residuals, and tropospheric errors, tailored specifically to 5G NTN signals and assumptions. In addition, realistic SNR characterizations derived from representative 3GPP TR 38.821 study cases are integrated into the model to evaluate error components under practical operating conditions.
- 2) *Constellation Geometry and Dilution of Precision (DOP) Assessment*: A DOP framework is employed to quantify the impact of satellite geometry on positioning accuracy, considering representative LEO constellations at 600 km and 1200 km altitudes.
- 3) *Minimum Satellite Visibility Analysis*: A dedicated evaluation of satellite visibility probabilities is conducted for the LEO-600 and LEO-1200 constellations, quantifying the likelihood of observing at least four satellites above different elevation masks. These results provide key insights into coverage availability and positioning service continuity.
- 4) *Performance Evaluation for PRS and SSB*: Through Monte Carlo (MC) simulations, the 95th-percentile horizontal positioning errors for both PRS-based and SSB-based positioning are evaluated across different deployment scenarios. The results demonstrate the superior accuracy of PRS signals and highlight the viability of SSB-based positioning with moderate trade-offs.

The remainder of this paper is organized as follows. Section II formulates the UERE and details the modeling of its individual components. Section III introduces the DOP framework and the approach for deriving positioning accuracy. Section IV describes the simulation scenarios, including satellite constellation parameters, link budget configurations, PRS settings, and MC evaluation methodology, and presents the corresponding GDOP and positioning accuracy results. Finally, Section V concludes the paper and discusses perspectives for future enhancements and research directions.

II. USER EQUIVALENT RANGE ERROR FORMULATION

The UERE is a fundamental metric for assessing the overall error affecting pseudorange measurements in positioning systems. It quantifies the cumulative impact of various independent error sources that influence position estimation accuracy [28]. Widely applied in satellite-based navigation systems, including GNSS and emerging LEO-PNT, UERE provides a structured framework for analyzing and mitigating positioning inaccuracies. A comprehensive understanding of

UERE components enables engineers to enhance positioning accuracy and improve the reliability of location-based services.

The total UERE is computed by aggregating the contributions of multiple error sources, categorized into three primary components: Satellite-Related Errors, Receiver-Related Errors, and Propagation-Related Errors.

A. SATELLITE-RELATED ERRORS

Satellite-related errors constitute a critical factor in positioning accuracy, originating from imperfections in the satellite's onboard systems. These errors primarily stem from clock deviations and ephemeris inaccuracies, both of which directly affect the precision of navigation solutions.

The following subsections provide a detailed examination of these error sources.

Satellite Clock Errors (σ_{Clk}): It arises from inaccuracies in the onboard timing systems of satellites, which play a critical role in ensuring precise positioning calculations. The performance of any positioning technique based on time measurements is highly dependent on the accuracy and stability of the time reference used by both the satellite and the user terminal. For example, the GPS Time (GPST) [29] serves as the standard time system for the Global Positioning System (GPS) and is maintained through a network of highly stable atomic clocks, both on the ground and onboard the satellites.

The landscape of satellite clock technologies is highly diverse, ranging from the ultra-stable atomic clocks [28, Sec. 2.7] used in MEO GNSS constellations, to the comparatively less stable oscillators found in LEO CubeSats [30]. While some LEO satellites, such as those used in radio occultation missions, employ Ultra-Stable Oscillators (USO), the majority of LEO CubeSats rely on Oven Controlled Crystal Oscillator (OCXO) or, in some cases, the less stable Temperature-Compensated Crystal Oscillator (TCXO), which tend to exhibit significant drift over time.

In the context of 5G NTN, 3GPP has not yet established explicit requirements regarding the minimum satellite clock performance necessary for positioning. The only related requirements are those specified in 3GPP TS 38.104 [13, Sec. 6.5.1.2], which define the frequency accuracy for gNB: ± 0.05 ppm for wide-area base stations and ± 0.1 ppm for medium-range and local-area base stations over a 1 ms period. However, these frequency stability requirements are designed primarily for communication purposes and cannot serve as a baseline for positioning applications. In fact, an oscillator with this level of precision would introduce significant timing errors, resulting in up to 30 meters of ranging error per second for a stability of ± 0.1 ppm, thereby making ToA-based positioning techniques impractical without compensation mechanisms.

For this study, a GNSS Disciplined Oscillator (GNSSDO) is considered as the baseline time reference [31]. A GNSSDO combines the long-term stability of timing information derived from a GNSS time reference with the short-term

stability of an internal high-stability oscillator, typically an OCXO. This configuration provides a cost-effective, stable, and reliable time and frequency reference. GNSSDOs are increasingly used in various terrestrial measurement applications [32] as well as in space missions, such as the Sentinel-6A Earth observation satellite of the Copernicus program [33].

For the purpose of numerical evaluation, we consider a representative example of a multi-frequency, multi-constellation GNSS receiver equipped with an OCXO-disciplined oscillator. As an illustration, the NovAtel OEM719 receiver, often used in space-related applications, specifies a timing Root Mean Square (RMS) error of approximately 5 ns [34]. Based on this reference value, the total satellite clock error budget, denoted by σ_{Clk} , is assumed to be $\sigma_{Clk} = 5$ ns. Other potential sources of timing error, such as short-term oscillator stability or Phase-Locked Loop (PLL) jitter, are neglected in this study.

Satellite Ephemeris Errors (σ_{Eph}): Accurate satellite positioning is essential for LEO-PNT systems, regardless of the positioning technique used. Unlike MEO and Geostationary Orbit (GEO) satellites, LEO satellites operate much closer to Earth and are significantly affected by gravitational perturbations, atmospheric drag, and other dynamic forces. These effects introduce rapid orbital variations, making traditional GNSS ephemeris models less effective for LEO satellites [35]. Consequently, ongoing research is focused on developing new modeling approaches to ensure precise satellite positioning in the LEO context. A comprehensive literature review is provided in [36].

Two main approaches can be considered for generating and broadcasting LEO satellite positions to UE:

- **Orbit Prediction Model:** Similar to GNSS, this method transmits an orbit prediction model within the navigation message, allowing the UE to determine the satellite's position over a prediction period. In practice, the orbital prediction model is constructed from a set of Keplerian orbital parameters, possibly augmented with correction coefficients that account for perturbations such as gravitational anomalies, atmospheric drag, and solar radiation pressure. The broadcast message thus contains a compact representation of the satellite's trajectory, which the UE propagates forward in time using orbital dynamics equations. For instance, the authors in [37] developed a broadcast model using 22 Keplerian parameters, compared to the 16 used in legacy GNSS navigation messages. With 16 parameters, the user range error for LEO satellites at an altitude of 800 km ranged from approximately 4 to 18 m, while incorporating additional coefficients significantly improved the accuracy down to the centimeter level. This predictive approach reduces the need for continuous ephemeris updates, while still providing sufficient accuracy for reliable positioning services.
- **Instantaneous Satellite Coordinates:** This method directly provides the satellite's position at the exact

epoch of transmission. The user equipment must demodulate the satellite position message whenever it performs a pseudorange measurement. The recent multi-constellation, multi-frequency GNSS receivers can achieve standalone (i.e., without augmentation system) positioning accuracy as low as 1 meter under favorable conditions [38], [39]. A GNSS receiver combined with an augmentation system can achieve decimetric accuracy, such as the NaviLEO receiver developed by SpacePNT SA [40].

For now, the 3GPP 5G NTN framework does not specify explicit requirements for satellite positioning accuracy. However, precise satellite position data is crucial for achieving reliable positioning services. Additionally, 5G NTN systems currently disseminate satellite ephemeris information through the System Information Block 19 (SIB19) broadcast message, which includes:

- Cartesian satellite position and velocity for real-time tracking.
- Six Keplerian parameters for orbit characterization.

The present configuration of SIB19 may pose challenges for high-precision 5G NTN positioning services for both types of disseminated information. For Cartesian satellite position, 3GPP TS 38.331 (Release 18) specifies a step granularity of $\Delta_{SIB19} = 1.3$ m for each coordinate [41, p. 792], which constitutes a limiting factor even when using a decimeter-level onboard GNSS receiver. According to [42, p. 196], this quantization step results in a one-dimensional RMS error of

$$\varepsilon_{\text{RMS,1D}} = \frac{\Delta_{\text{SIB19}}}{\sqrt{12}} \approx \frac{1.3}{\sqrt{12}} \approx 0.375 \text{ m} \quad (1)$$

and a three-dimensional RMS error of

$$\varepsilon_{\text{RMS,3D}} = \sqrt{3} \cdot \varepsilon_{\text{RMS,1D}} \approx 0.65 \text{ m} \quad (2)$$

Regarding the disseminated six Keplerian parameters in the SIB19, they typically allow for satellite position accuracy in the order of a few kilometers. Such accuracy is insufficient for precise standalone UE-based positioning and is only suitable for network-verified UE location applications, as introduced in the study item TR 38.882 [12].

As a baseline for this study, we assume the instantaneous satellite coordinates method using the NovAtel OEM719 receiver. This multi-frequency GNSS receiver, widely used in space applications, achieves an RMS position error of 1.2 meters in dual-frequency mode, which is equivalent to a time-domain error of $\sigma_{Eph} = 4$ ns.

B. RECEIVER-RELATED ERRORS

Receiver-related errors arise from imperfections in the user equipment used to process satellite signals. The following subsections describe these errors in detail.

Time of arrival (ToA) measurement error (σ_{ToA}): In time-based positioning receivers, the ToA of the signal is estimated by applying a matched filter (i.e., correlator) to the incoming signal, using a known pseudorandom code [43].

This process maximizes the SNR and enables estimation of both the signal's time delay and Doppler shift [44]. In practice, most receivers employ a Delay Locked Loop (DLL) to track continuously the code phase of the incoming signal [28]. Two principal error sources affect the DLL tracking performance:

- *Thermal Noise Error* σ_t : Thermal noise imposes a fundamental limit on the accuracy of ToA estimation. A convenient lower bound on the standard deviation of the ToA estimate is given by the Cramer-Rao Lower Bound (CRLB) [45, Sec. 3.4]. According to the general CRLB derivation for the estimation of time delay of signals in Gaussian white noise, as given by Kay [45, p. 53], the variance of the estimated time delay $\hat{\tau}$ is bounded by:

$$\text{var}(\hat{\tau}) \geq \sigma_{\text{CRLB}}^2 = \frac{1}{\frac{E_s}{N_0/2} \cdot \bar{F}^2} \quad (3)$$

where:

- E_s is the total accumulated energy over one Orthogonal Frequency-Division Multiplexing (OFDM) symbol with $E_s = C/\Delta_f$, C is defined as the power of the band-pass signal and Δ_f is the subcarrier spacing,
- N_0 is the noise power spectral density,
- $\bar{F}^2$ is the Mean-Squared Bandwidth (MSB) of the signal.

The MSB can be seen as a measure of the sharpness of the correlation peak [28]. Clearly, signals with larger MSB offer the potential for more accurate code tracking. It can be approximated for OFDM symbol of N subcarriers as follows [46]:

$$\bar{F}^2 \approx 4\pi^2 \frac{\Delta_f^2}{N} \sum_{k \in \mathcal{N}_a} p_k^2 \cdot k^2 \quad (4)$$

where:

- p_k^2 denotes the relative power weight of subcarrier k , satisfying $\sum_k p_k^2 = N$.
- $\mathcal{N}_a$ denotes the set of N_a active pilot subcarrier indices, arranged symmetrically around the DC component, and spanning the range from $-N/2$ to $+N/2 - 1$.

Assuming that the power is evenly distributed across all subcarriers carrying PRS, the relative power weight of a subcarrier k is:

$$p_k = \sqrt{\frac{N}{N_a}} = \sqrt{K} \quad (5)$$

where K is the comb size parameter, which defines the spacing between PRS subcarriers. The number of subcarriers N and allowed comb values are specified in TS 38.211 [47], with $K \in \{2, 4, 6, 12\}$.

The value of $\bar{F}_{\text{PRS}}^2$ can be recalculated by expanding (4), applying (5), and assuming a large value of N , as follows:

$$\bar{F}_{\text{PRS}}^2 \approx \frac{\pi^2 \Delta_f^2 N^2}{3} \quad (6)$$

Therefore, the CRLB for the delay estimation based on M PRS symbols is given by:

$$\sigma_{\text{CRLB, PRS}}^2 = \frac{3}{2\pi^2 \Delta_f^2 M N^2 \text{SNR}} \quad (7)$$

Assuming that the ToA is estimated using a DLL updated every integration time T_{Int} , the ToA measurement errors $\sigma_{\text{CRLB, PRS}}$ are filtered by a loop filter with noise bandwidth B_n , independently of the discriminator architecture. The variance of the filtered ToA, denoted $\sigma_{t, \text{PRS}}^2$, can be derived as in [48] and [49], as follows:

$$\begin{aligned} \text{var}(\hat{\tau}_{\text{PRS}}) &\geq \sigma_{t, \text{PRS}}^2 \\ \sigma_{t, \text{PRS}}^2 &= 2\sigma_{\text{CRLB, PRS}}^2 B_n T_{\text{Int}} (1 - 0.5 B_n T_{\text{Int}}) \\ &= \frac{3B_n T_{\text{Int}} (1 - 0.5 B_n T_{\text{Int}})}{\pi^2 \Delta_f^2 M N^2 \text{SNR}} \end{aligned} \quad (8)$$

In all previous analytical derivations, it is assumed that the PRS is continuously transmitted and received, as is the case for GNSS signals. However, in practice, the PRS is only present for a fraction of the coherent integration time T_{Int} . This fraction, denoted by γ , can be expressed as

$$\gamma = \frac{M}{\Delta_f \cdot T_{\text{Int}}} \quad (9)$$

For instance, considering a PRS configuration with 12 symbols per two 5G frames in FR1, and a subcarrier spacing of $\Delta_f = 30$ kHz, the PRS occupies approximately $\gamma = 4\%$ of the duration of the integration time $T_{\text{Int}} = 20$ ms. Consequently, SNR used in (8) corresponds to the effective SNR, which is related to the injected SNR, denoted by SNR_{inj} , and defined as [50]:

$$\text{SNR} = \gamma \text{SNR}_{\text{inj}} \quad (10)$$

By substituting (10) and (9) into (8), the final expression for the ToA variance is obtained as follows:

$$\sigma_{t, \text{PRS}}^2 = \frac{3B_n T_{\text{Int}}^2 (1 - 0.5 B_n T_{\text{Int}})}{\pi^2 \Delta_f^2 M^2 N^2 \text{SNR}_{\text{inj}}} \quad (11)$$

The same reasoning applied for the PRS can also be used for the SSB, leading to the following analytical expressions. The CRLB for the SSB is given by:

$$\sigma_{\text{CRLB, SSB}}^2 = \frac{3}{2\pi^2 \Delta_f^2 M (N^2 + 2) \text{SNR}} \quad (12)$$

Assuming that the ToA is estimated using a DLL, the variance of the filtered ToA estimate for the SSB, denoted by $\sigma_{t, \text{SSB}}^2$, is given by:

$$\sigma_{t, \text{SSB}}^2 = \frac{3B_n T_{\text{Int}}^2 (1 - 0.5 B_n T_{\text{Int}})}{\pi^2 \Delta_f^2 M^2 (N^2 + 2) \text{SNR}_{\text{inj}}} \quad (13)$$

In the case of the SSB, the number of subcarriers N and the number of OFDM symbols M are fixed, with $N = 240$

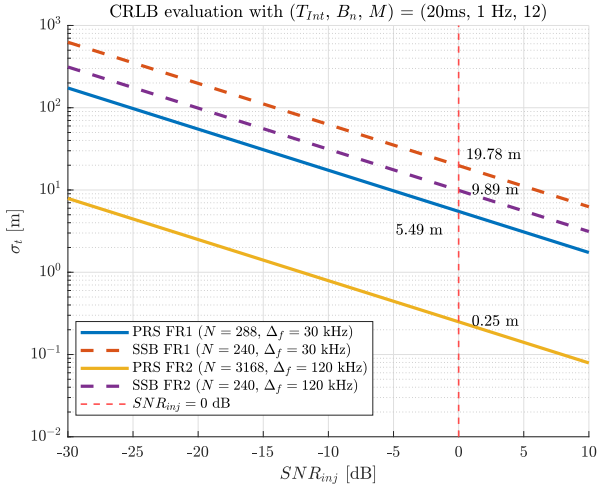


FIGURE 1. The CRLB-based evaluation of thermal noise-induced ToA error for one PRS slot of 12 symbols and for SSB configurations in FR1 and FR2 scenarios, with $T_{\text{int}} = 20$ ms and $B_n = 1$ Hz. The corresponding timing error standard deviations σ_t at $\text{SNR}_{\text{inj}} = 0$ dB are also indicated.

and $M = 4$. Using these fixed values, the CRLB for the SSB is given by:

$$\sigma_{t,\text{SSB}}^2 = \frac{3B_n T_{\text{int}}^2 (1 - 0.5B_n T_{\text{int}})}{921632\pi^2 \Delta_f \text{SNR}_{\text{inj}}} \quad (14)$$

Figure 1 presents an example evaluation of the expressions in (11) and (14) for two typical 5G NTN positioning scenarios: FR1 and FR2. In the FR1 case, the subcarrier spacing Δ_f was set to 30 kHz, and the signal bandwidth was approximately 10 MHz. For FR2, a subcarrier spacing of 120 kHz was assumed, resulting in a signal bandwidth of about 400 MHz. In both scenarios, the PRS was configured with $M = 12$ symbols per two 5G frame (i.e., every 20 ms). It is evident that the thermal noise error is inversely proportional to both the SNR and the signal bandwidth. The PRS configurations exhibit lower errors compared to their SSB counterparts due to their denser time-frequency occupancy. The PRS in FR1 demonstrates better positioning accuracy than the SSB in FR2, despite its smaller bandwidth. This is attributed to the significantly longer time occupancy of the PRS signal ($12 \times 1/30$ kHz = 0.4 ms) compared to the SSB ($4 \times 1/120$ kHz = 0.033 ms), which results in higher integration time and improved robustness to thermal noise. In FR2, the significantly higher bandwidth available to the PRS (380.2 MHz) enables a drastic reduction in σ_t , achieving sub-meter accuracy at SNR levels as low as -12 dB. These results underline the critical role of signal structure and bandwidth in achieving high-precision ToA estimation.

- **Dynamic Stress Error σ_d :** The second source of error when estimating the ToA using a DLL arises from the dynamic stress due to satellite or receiver motion (acceleration, jerk, etc.). According to [28, p. 492], the loop dynamic stress error is given by:

$$\sigma_d = \frac{1}{c} \frac{d^n \rho}{\omega_0^n} \quad (15)$$

where

- ρ is the code range observable (in meters),
- $\frac{d^n \rho}{dt^n}$ is the n -th time derivative of ρ (e.g., acceleration for $n = 2$, jerk for $n = 3$),
- ω_0 is the loop filter natural radian frequency of the n -th-order DLL, determined from the loop filter noise bandwidth B_n .
- c is the speed of light.

For example, consider a *second-order* DLL ($n = 2$) with a noise bandwidth $B_n = 1$ Hz. It is common to take $B_n \simeq 0.53 \omega_0$ [51]. Assuming a maximum Line of Sight (LOS) jerk of 1 m/s^3 (typical for a LEO satellite at 600 km), the resulting maximum dynamic stress error is $\sigma_d \approx 1.76$ ns.

For comparison, using the same loop conditions for an MEO GNSS satellite, where the maximum LOS jerk is only $\approx 0.000122 \text{ m/s}^3$, the corresponding dynamic stress error is a negligible $\sigma_d \approx 0.21$ ps.

- **Total ToA measurement error σ_{ToA} :** The total ToA error results from the combination of thermal noise and dynamic stress-related errors, expressed as

$$\sigma_{\text{ToA}} = \sqrt{\sigma_t^2 + \sigma_d^2} \quad (16)$$

However, the use of a carrier-aided DLL (practically a universal practice) effectively removes the dynamic stress error [28, p. 488]. Consequently, we assume that the receiver employs carrier-aided code tracking, making the dynamic stress effect negligible in subsequent analyses. Thus, we take $\sigma_{\text{ToA}} = \sigma_t$.

Multipath Errors (σ_{Multi}): Multipath refers to the reception of the signal of interest along with several copies of it. These copies are reflections of the signal on nearby elements, such as the ground, buildings, etc. Each of these copies has a relative delay τ_d with the main signal and a relative power α_d . The maximum delay of a reflection relative to the main signal is known as Delay Spread (DS). In scenarios where there is a clear LOS, the relative power of each multipath is usually modeled as a Rice distribution, with a parameter $\mathcal{K}$ called the Rice factor. The literature reports a variety of models that follow this approach [52], [52], [53], [54], [55], [56]. Nevertheless, for LEO satellite communications, only a limited number of channel models explicitly incorporate the satellite elevation angle as a parameter [57], [58], although this dependency is critical for PNT applications due to its direct impact on the GDOP. Accordingly, in this work we adopt the recently published model described in [58], Section IV-B, for the C-band. The model developed in [58] employs a ray-tracing technique to derive the channel characteristics for LEO satellites, explicitly accounting for the dependency on the satellite elevation angle. This model provides detailed parameterizations of multipath effects as a function of the elevation angle under different environmental scenarios. Since the model parameters are extracted only for discrete elevation angles, we interpolate the values between two consecutive elevation angles.

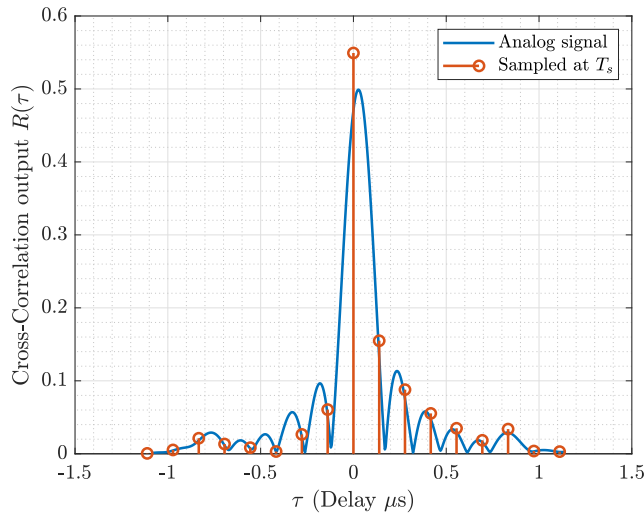


FIGURE 2. The two-path effect appears in the correlation output when the relative delay between the signals is smaller than the sampling interval. The resulting multipath error arises from two main factors: (i) the peak position of the analog signal differs from that of the discretized signal, and (ii) the broadening of the main correlation lobe. This error can be mitigated by employing larger bandwidths, which increase the resolution of the discrete signal and can also decompose the wide main lobe into narrower lobes, one corresponding to each path.

In this context, the multipath model is typically discretized according to the propagation environment, such as dense urban, urban, suburban, or rural, each characterized by different building and tree densities and heights, which are the main contributors to multipath. Accordingly, the multipath environment is described by the number of signal paths M , the maximum delay spread $\tau_{\max}$, and the Rice factor K for each scenario. The work in [58] provides parameter models for dense urban, urban, and suburban environments, which are adopted in this study.

Multipath interference influences the range estimation due to the superposition of reflected signals at the receiver's correlation output. Instead of a single distinct peak, reflections may produce additional peaks that merge or closely follow the primary path, complicating the isolation of the true first-arrival signal. Figure 2 illustrates the resulting range error in a simplified example with two closely spaced signal paths when the signal bandwidth is low (using 10 Resource Block (RB)). Here, the delay spread between them is shorter than the sampling interval, thereby increasing the width of the primary lobe of the cross correlation that correspond to the first-arrival signal. While in Figure 3 the larger bandwidth of the signal (80 RB) makes the lobes narrower, thus, it is possible now to estimate the second path from the correlation output.

To derive a practical multipath error model that captures the dependency on the satellite elevation angle, we perform a MC simulation. This simulation evaluates the variability of range estimation errors under different satellite elevation angles θ . Specifically, the simulation generates the PRS signal, applies the channel, and estimates the delay using cross-correlation as a matched filter. The simulator output

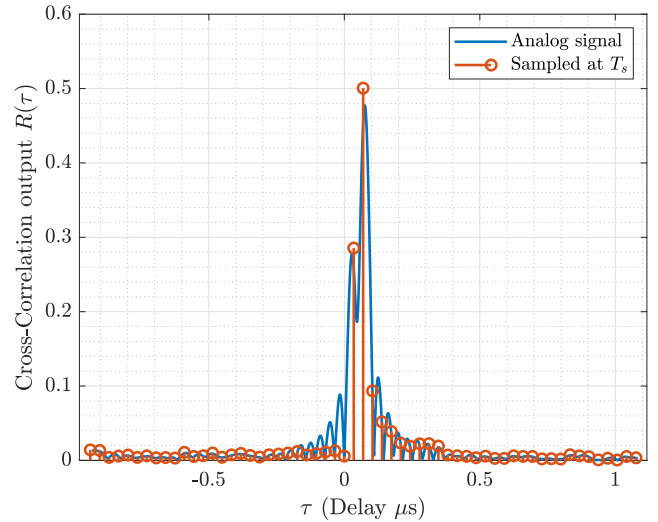


FIGURE 3. The two-path effect in the correlation output becomes evident when the relative delay exceeds the sampling interval, in which case the two received paths are clearly distinguishable.

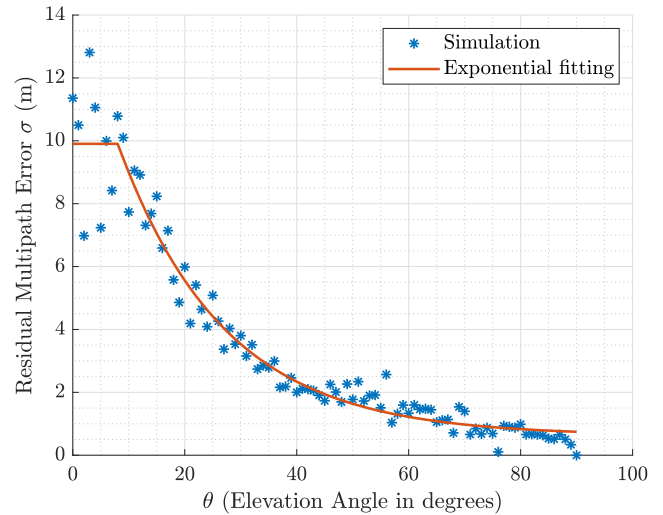


FIGURE 4. Multipath range error fitting for the Urban scenario.

is the variance of the delay estimation for a given elevation angle.

The PRS configuration adopted corresponds to the densest occupation of the OFDM grid, namely 12 PRS symbols per slot with a comb size of 2, a subcarrier spacing of 30 kHz, and 12 RB.

After running the simulation, we found that the multipath error remains constant up to a threshold elevation angle θ_{th} , after which the error decays exponentially. Figure 4 is an example of the simulation output for the scenario “Urban”.

The model in Figure 4 follows

$$\sigma_{Multi}(\theta) = \begin{cases} A, & \theta \in [0, \theta_{th}] \\ B + Ce^{-\theta/D}, & \theta \in (\theta_{th}, \pi/2], \end{cases} \quad (17)$$

where the parameters θ_{th} , A , B , C , and D vary according to the scenario under consideration, such as Deep Urban, Urban or Suburban.

TABLE 2. Multipath model parameters for different scenarios.

Scenario	θ_{th}	A	B	C	D
Deep Urban	30°	12.2	-1.42	12.2	24.5
Urban	10°	9.92	0.62	14.2	18.9
Suburban	0°	5.36	0.50	6.91	20.1

The values of θ_{th} , A , B , C , and D are obtained empirically by using a least-squares minimization from the results of 1000 MC iterations per each elevation angle of the satellite and per each channel model. For each dataset we have extracted the $\sigma_{Multi}(\theta)$ and fit these $\sigma_{Multi}(\theta)$ to (17). Finally, Table 2 summarizes the parameters found that fit σ_{Multi} to (17) for each scenario.

This elevation-dependent multipath error model, obtained through MC simulations and exponential curve fitting, serves as the baseline for deriving the multipath error component in the overall UERE budget. Specifically, the fitted function shown in Figure 4 is incorporated into the final error computation to capture the impact of both environmental conditions and satellite elevation angle on multipath propagation.

C. PROPAGATION-RELATED ERRORS

Propagation-related errors occur when satellite signals travel through the Earth's atmosphere, causing inaccuracies in measuring the distances between satellites and receivers. The following subsections provide a detailed explanation of these errors.

Residual Ionospheric Delay (σ_{Iono}): The ionosphere is a dispersive medium located primarily between about 70 km and 1,000 km above the Earth's surface. In this region, solar ultraviolet radiation ionizes a portion of the atmospheric gas molecules, releasing free electrons. These free electrons affect the propagation of electromagnetic waves, including 5G NTN satellite broadcasts. Because the ionosphere is dispersive, the propagation velocity v_p of the signal's carrier phase differs from the group velocity v_g that carries the signal information. The amount of group-velocity retardation is equal to the carrier-phase advance relative to free-space propagation [59, Sec. 6.3.2].

The propagation velocity depends on the refractive index, which in turn is related to the electron density along the satellite-to-user signal path. This electron density is referred to as the total electron content Total Electron Content (TEC) and is expressed in units of electrons/m². The TEC depends on the time of day, user location, satellite elevation angle, season, ionizing flux, magnetic activity, sunspot cycle, and scintillation. It typically ranges from about 10¹⁶ to 10¹⁹, with the lower end around midnight and the upper end around mid-afternoon.

For positioning based on code-phase measurements (e.g., using the 5G PRS signal), the code experiences a delay in seconds given by [28, p. 637]:

$$\Delta S_g^{Iono} = \frac{40.3 \times 10^6 F_{pp}(\theta) \text{ TVEC}}{cf^2} \quad (18)$$

where:

- Vertical Total Electron Content (VTEC) is the vertical TEC expressed in TEC Units (TECU), typically provided by ionospheric models,
- F_{pp} is an obliquity (mapping) factor for a given elevation angle θ , accounting for the increased path length through the ionosphere compared to the vertical (zenith) path computed;
- f is the carrier frequency,
- c is the speed of light.

From (18), the ionospheric delay is inversely proportional to the square of the frequency and proportional to the total number of free electrons along the path. Consequently, the effect of ionospheric delay on positioning accuracy (i.e., the total error budget) strongly depends on the signal frequency for the same TEC. In GNSS frequency bands, delay errors on the order of tens of meters can be expected [60]. In the K-band, these errors remain well below 1 m and can thus be neglected [61], [62]. Therefore, for 5G NTN FR2 (at $f \geq 20$ GHz), the ionospheric errors are assumed to be negligible ($\sigma_{Iono} \approx 0$ ns).

For FR1, the ionospheric delay is expected to be non-negligible, requiring a correction. The first method to correct for this delay takes advantage of its frequency dependence; it can be virtually eliminated by making ranging measurements with a dual-frequency receiver. Differencing pseudorange measurements made on two frequencies enables the estimation of the delays on both frequencies, neglecting multipath and receiver noise errors. This method is already widely used in modern consumer electronics and professional devices [63].

The second method consists of using ionospheric models to estimate the VTEC and then the ionospheric delay using equation (18). As of today, there is no dedicated ionospheric model specifically designed for LEO satellites in PNT applications. However, for MEO GNSS constellations, several ionospheric models are widely used in real-time PNT applications, such as:

- The Klobuchar model [64], currently employed by GPS;
- The NeQuick-G model [65], used by Galileo;
- The BeiDou Global Broadcast Ionospheric Delay Correction Model [66].

Since these models were originally designed for MEO, they cover the entire ionosphere. Consequently, the same ionospheric model can theoretically be adapted to the LEO scenario. The ionospheric delay experienced in LEO is expected to be lower than in MEO [67].

There is not yet an established adaptation of these models despite recent efforts toward LEO applications, such as those presented in [61]. For this study, we approximate the expected residual ionospheric error using the NeQuick-G model, as it outperforms the Klobuchar model [68] and removes at least 70% of the ionospheric error [69]. The residual ionospheric error must be scaled from GNSS L1 at $f_{L1} = 1.57542$ GHz to the FR1 frequencies as

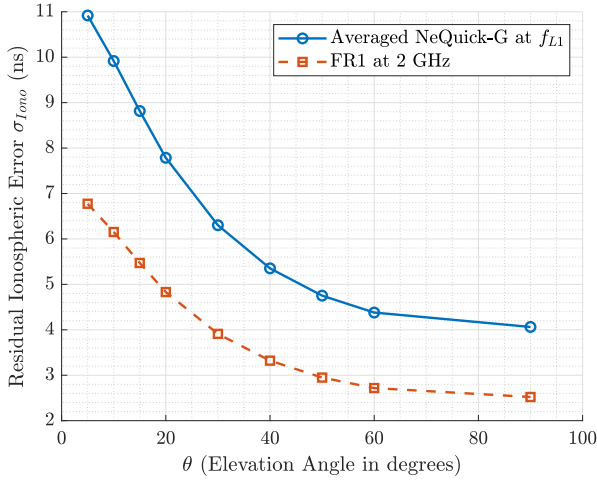


FIGURE 5. NeQuick-G baseline residual ionospheric error for f_{L1} and f_{FR1}^* .

follows [67, p. 52]:

$$\sigma_{Iono}^{FR1} = \left(\frac{f_{L1}}{f_{FR1}} \right)^2 \sigma_{Iono}^{L1} \quad (19)$$

The baseline for the residual ionospheric error of the NeQuick-G model is derived from two years of global monitoring data, averaged worldwide [70]. The baseline values of NeQuick-G, along with the scaled values for FR1, are presented in Figure 5.

The scaled values for FR1 shown in Figure 5 are adopted in this study as the baseline for modeling the residual ionospheric error component of the overall UERE budget.

Residual Tropospheric Delay (σ_{Tropo}): The troposphere is the atmospheric layer extending from the Earth's surface up to the ionosphere. Due to its non-dispersive nature, the troposphere introduces a frequency-independent delay, known as a non-frequency-selective delay, to propagating signals. Consequently, both phase and group velocities experience an identical delay relative to free-space propagation. This delay is governed by the refractive index of the troposphere, which depends primarily on local atmospheric conditions such as temperature, pressure, and relative humidity.

Tropospheric delays are conventionally modeled by separating them into dry and wet components [71]. The dry component is attributed to dry gases present in the atmosphere, while the wet component arises from atmospheric water vapor. The dry component, which can be accurately predicted, represents approximately 90% of the tropospheric delay [72]. The total tropospheric delay in meters is typically modeled as:

$$\tau_{Tropo} = m_{dry}(\theta) \tau_{dry} + m_{wet}(\theta) \tau_{wet} \quad (20)$$

where:

- $m_{dry}(\theta)$ and $m_{wet}(\theta)$ are the mapping functions (or obliquity factors) corresponding to dry and wet components, respectively, as functions of satellite elevation angle θ ,

- τ_{dry} and τ_{wet} represent the zenith tropospheric delays associated with the dry and wet atmospheric components, respectively.

The mapping functions transform zenith delays into slant delays specific to each satellite signal path. By employing a standardized zenith delay model alongside appropriate mapping functions, accurate slant delay predictions can be obtained for various receiver locations and satellite geometries. Several zenith delay models exist, including the Saastamoinen model [73] and the Hopfield model [74]. These models require various meteorological parameters, such as atmospheric pressure and temperature, to estimate the zenith delay accurately.

Similarly, multiple mapping functions have been proposed [75], including the Herring mapping function and the Niell mapping function. The study in [67] compared various combinations of zenith delay models and mapping functions, finding that most combinations yield similar results, with the notable exception of Hopfield's model, which exhibited greater deviations.

In this study, as a baseline for the analysis, we utilize the RTCA simplified model (originally developed by the Radio Technical Committee for Aeronautics) [76]. This model is straightforward and does not require real-time meteorological data. It assumes a standard atmosphere with a fixed zenith delay value of $\sigma_{TVE} = 0.12$ m, and the slant delay is then computed using the mapping function $m(\theta)$:

$$\sigma_{Tropo} = \sigma_{TVE} \cdot m(\theta) \quad (21)$$

where the mapping function for an elevation angle θ is defined as:

$$m(\theta) = \frac{1.001}{\sqrt{0.002001 + \sin^2 \theta}} \quad (22)$$

Figure 6 presents the baseline values for the residual tropospheric error according to (21). Note that this model is considered conservative [77], and the use of a more sophisticated model incorporating meteorological parameters is expected to reduce this residual tropospheric delay.

The values shown in Figure 6 are adopted in this study as the baseline model for the residual tropospheric delay component within the overall UERE budget.

D. TOTAL UERE BUDGET

The UERE for the satellite i is computed as the root sum square of all error components, reflecting the overall uncertainty in range estimation:

$$\sigma_{UERE,i} = \left(\sigma_{Ck,i}^2 + \sigma_{Eph,i}^2 + \sigma_{ToA,i}^2 + \sigma_{Multi,i}^2 + \sigma_{Iono,i}^2 + \sigma_{Tropo,i}^2 \right)^{\frac{1}{2}} \quad (23)$$

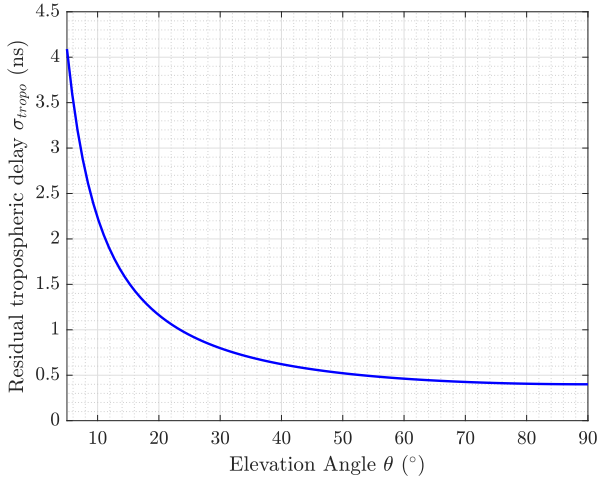


FIGURE 6. Residual tropospheric delay as a function of elevation Angle.

III. DILUTION OF PRECISION MODEL AND POSITIONING ACCURACY

The receiver's actual position is determined iteratively, starting from an initial estimate of the receiver's position [28, Sec. 2.5.2]. This initial estimate is referred to as the linearization point. The vector ΔX , which represents the user's position and clock bias relative to the linearization point, is related to the pseudo-range residuals $\Delta\rho$ by the following equation [78]:

$$\Delta\rho_{N \times 1} = H_{N \times 4} \Delta X_{4 \times 1} + \epsilon_{N \times 1} \quad (24)$$

where:

- $H_{N \times 4}$ is the design matrix that relates the measurements to the estimated parameters;
- ϵ is the measurement error vector, where $\epsilon_{N \times 1} \sim \mathcal{N}(0, Q_{N \times N})$.
- $Q_{N \times N}$ is the covariance matrix of the measurement errors;
- N is the number of visible satellites.

The pseudorange residuals $\Delta\rho$ are assumed to be affected by independent Gaussian noise with zero mean and a heteroscedastic covariance matrix Q . This assumption is justified by the fact that the UEREs of visible satellites are not identically distributed. Satellites with higher elevation angles are expected to exhibit lower UEREs. Therefore, the matrix Q is defined as follows [79]:

$$Q = \begin{bmatrix} \sigma_{UERE_1} & 0 & 0 & \cdots & 0 \\ 0 & \sigma_{UERE_2} & 0 & \cdots & 0 \\ 0 & 0 & \sigma_{UERE_3} & \cdots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & \cdots & \sigma_{UERE_N} \end{bmatrix} \quad (25)$$

When $N \geq 4$, the optimal estimation of the user's position is obtained through the weighted least squares method, given by [28, p. 941]:

$$\Delta X = (H^T Q^{-1} H)^{-1} H^T Q^{-1} \Delta\rho \quad (26)$$

The error covariance matrix of ΔX is derived after assuming that the displacement vector ΔX is corrupted by errors following a zero mean Gaussian distribution as follows:

$$\text{cov}(\Delta X) = \mathbb{E}[\Delta X \Delta X^T] = (H^T Q^{-1} H)^{-1} \quad (27)$$

The classical definition of GDOP (which is dimensionless) assumes that all measurements share the same noise variance, so that the measurement covariance can be factored out as a single scalar. However, if the measurement-error covariance is not simply $\sigma^2 \mathbf{I}$, the classical GDOP no longer applies. Consequently, we adopt the Weighted Dilution of Precision (WDOP) [79], which accounts for both the satellite geometry and the measurement-error covariance. It is given by

$$\text{WDOP} = \text{cov}(\Delta X) = (H^T Q^{-1} H)^{-1} \quad (28)$$

The accuracy of the estimated position at the 95% confidence interval, R_{95} , is approximated by [28, p. 675]:

$$R_{95} \approx 2 \times \text{WDOP} \quad (29)$$

To analyze the geometry of the assumed satellite constellation without considering measurement noise errors, we use the classical formula for GDOP:

$$\text{GDOP} = (H^T H)^{-1} \quad (30)$$

IV. SIMULATION-BASED RESULTS

The following results are based on the error models previously described. The simulated pseudorange is impaired using the UERE formulation, which includes satellite, receiver, and propagation errors. These components are integrated via the covariance matrix Q in the weighted least squares algorithm (see Section III), accounting for clock bias, ephemeris errors, ToA error, multipath, and atmospheric delays.

A. MONTE CARLO SIMULATION DESCRIPTION

MC simulations with 100 000 trials were conducted to evaluate the expected positioning accuracy of 5G NTN systems using LEO satellite constellations. Satellite positions were generated over six orbital periods (approximately 600 minutes) for both LEO-600 and LEO-1200 configurations, using Keplerian orbital equations with a 15-second sampling interval.

In each trial, a random epoch was selected, and a receiver position was randomly generated, ensuring that at least four satellites were visible and the elevation angle satisfied $10^\circ < \theta < 90^\circ$. A latitude mask of $[-45^\circ, +45^\circ]$ was applied, encompassing the majority of the world population [80].

Each simulation computes the pseudorange errors using a detailed UERE formulation that includes satellite-related (clock, ephemeris), signal-related (ToA noise), and propagation-related (multipath, ionospheric, tropospheric) impairments. The resulting GDOP and positioning error at the 95% confidence level, R_{95} , are then evaluated to quantify

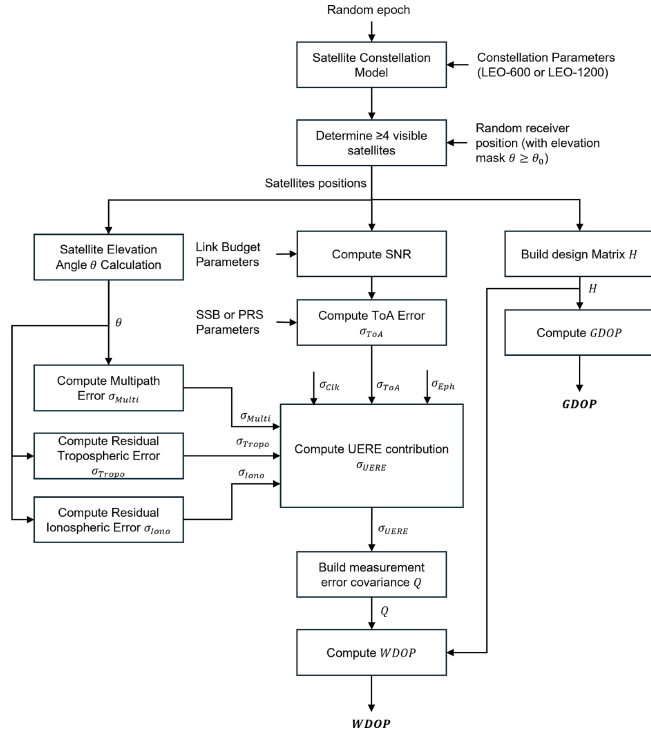


FIGURE 7. Simulation framework for UERE-based positioning performance evaluation.

the total expected positioning uncertainty under realistic conditions.

Figure 7 illustrates the complete simulation framework, including satellite geometry generation, link budget and error modeling, and computation of geometry- and measurement-dependent metrics.

B. SATELLITE CONSTELLATION

According to 3GPP TR 38.821 [81], different types of NTN platforms are classified based on their altitude range, from HAPS operating at 8–50 km to GEO satellites positioned at 35,786 km. Among these, LEO satellites, which operate between 350 km and 1,500 km, represent the emerging state-of-the-art due to their favorable link budget and lower latency [82], making them the focus of this study.

The orbital design of satellites significantly impacts system coverage. The key parameters in constellation design include orbit altitude, inclination angle, and constellation size [35]. 3GPP has defined reference NTN scenarios [81, p. 19], including circular LEO orbits at 600 km and 1,200 km, as well as GEO. This study focuses on the two LEO scenarios at 600 km and 1,200 km, referred to as LEO-600 and LEO-1200, respectively.

Currently, 3GPP has not yet specified all constellation parameters, such as the satellite orbital configurations and the total number of satellites. Therefore, to enable a consistent and representative simulation framework, we adopt the orbital configuration proposed in the 3GPP RAN1 contribution [83], which is based on circular polar orbits.

TABLE 3. Constellation parameters for LEO-600 and LEO-1200.

Constellation Name	LEO-600	LEO-1200
Satellite altitude (km)	600	1200
Orbit topology	Polar (87.5° inclination)	
Number of orbits	17	10
Number of satellites per orbit	30	17

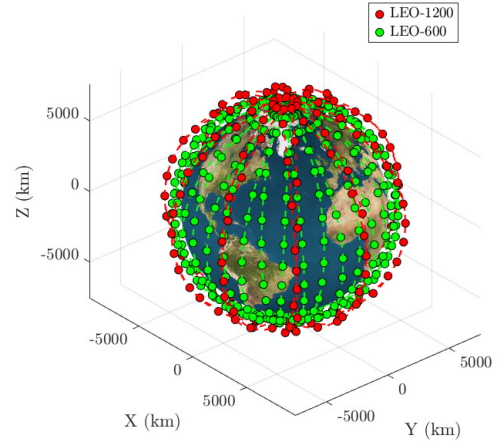


FIGURE 8. Illustration of the constellation baseline in ECEF coordinate system: LEO-600 with 17 orbital planes and LEO-1200 with 10 orbital planes.

These assumptions are fully aligned with current trends in commercial and governmental LEO deployments [82], [84]. The key parameters are summarized in Table 3. An illustration of these constellations LEO-600 and LEO-1200 is shown in Figure 8.

C. LINK BUDGET

The link budget is assessed based on the 5G NTN SNR calculations, as specified in [81, p. 53], for a transmission link between the satellite and the UE:

$$\text{SNR} = \text{EIRP}_d + \Gamma - k - P_{LFS} - P_{LA} - P_{LSM} - P_{LSL} - P_{LAD} \quad (31)$$

where:

- EIRP_d is the Equivalent Isotropic Radiated Power (EIRP) density (dBW/Hz), where $\text{EIRP}_d(\text{dB}) = \text{EIRP}(\text{dB}) - B(\text{dB})$,
- Γ is the receiver antenna gain-to-noise-temperature ratio (dB/K),
- k is Boltzmann's constant, which equals -228.6 dBW/K/Hz,
- P_{LFS} represents the free-space path loss (dB),
- P_{LA} denotes the atmospheric path loss due to gas absorption and rain attenuation (dB),
- P_{LSM} is the shadow fading margin (dB),
- P_{LSL} corresponds to the scintillation loss (dB),
- P_{LAD} accounts for additional losses, such as feeder link degradation in non-regenerative systems (dB).

TABLE 4. Detailed link budgets for representative 3GPP TR 38.821 study cases under different carrier frequencies, bandwidths, and orbital altitudes (LEO-600 and LEO-1200) [81, Table VI.1.3.3-1]. The SNR values are shown for an illustrative elevation angle of $\theta = 30^\circ$.

Constellation Name	LEO-600		LEO-1200	
3GPP Study Case Number	SC9	SC6	SC14	SC11
Carrier Frequency f_c (GHz)	FR1: 2	FR2: 20	FR1: 2	FR2: 20
Bandwidth B (MHz)	30	400	30	400
Satellite EIRP Density EIRP _d (dBW/MHz)	34	4	40	10
Receiver antenna gain-to-noise-temperature ratio Γ (dB/K)	-31.6	15.9	-31.6	15.9
Free space path loss $P_{L_{FS}}$ at 30° (dB)	159.1	179.1	164.5	184.5
Atmospheric loss P_{L_A} (dB)	0.1	0.5	0.1	0.5
Shadow fading margin $P_{L_{SM}}$ (dB)	3	0	3	0
Scintillation Loss $P_{L_{SL}}$ (dB)	2.2	0.3	2.2	0.3
Additional losses $P_{L_{AD}}$ (dB)	0	0	0	0
SNR (dB)	6.6	8.5	7.2	9.1

The parameters of equation (31) depend on the specific scenario under study. The 3GPP has defined 30 study cases in 3GPP TR 38.821 [81, Table 6.1.3.3-1]. To enhance clarity and streamline the analysis, we have selected four representative use cases that cover both FR1 and FR2 frequency ranges, as well as the two constellation types: LEO-600 and LEO-1200. The various parameters of equation (31) for each selected use case are summarized in Table 4.

The SNR values in Table 4 are directly derived from the reference link budgets defined in 3GPP TR 38.821 [81], which were contributed by industrial stakeholders (e.g., TAS, Nokia and Ericsson) and further documented in 3GPP RAN1 contribution R1-1913351 [85]. These values constitute the accepted reference conditions for NTN feasibility studies. While no explicit 3GPP specification mandates fixed SNR thresholds, the link budgets of TR 38.821 are widely used as the baseline for evaluating realistic operational conditions in LEO-based deployments.

D. POSITIONING REFERENCE SIGNAL CONFIGURATION

The downlink PRS is a dedicated signal defined in the 3GPP specifications to enable accurate downlink-based positioning. Its configuration is determined by a set of parameters described in TS 38.214 [86] and TS 38.211 [47]. In particular, the PRS periodicity can vary from 0.5 ms to 10.24 s, depending on the numerology index $\mu \in \{0, 1, 2, 3\}$. In this work, a periodicity of 20 ms is selected, as it is both supported for PRS and consistent with SSB transmission periodicities commonly adopted in practice.

To maximize the accuracy of the ToA estimation, the configuration assumes the highest PRS density, with a comb size of $K = 2$ and the maximum number of PRS symbols per slot ($M = 12$). For FR1, a subcarrier spacing of 30 kHz is adopted, whereas for FR2, 120 kHz is considered, following standard 5G numerologies.

In terms of occupied bandwidth, the SSB spans 7.2 MHz in FR1 (240×30 kHz) and 28.8 MHz in FR2 (240×120 kHz). For the PRS, bandwidths of approximately 30 MHz in FR1 and 400 MHz in FR2 are assumed, aligned with the representative NTN study cases defined in TR 38.821 (SC6, SC9, SC11, SC14). These choices ensure consistency with

standardized evaluation scenarios while enabling meaningful comparisons between PRS- and SSB-based positioning performance.

E. MINIMUM SATELLITE VISIBILITY ANALYSIS RESULTS

In this section, we evaluate the probability of observing at least four satellites above a given elevation mask θ_0 for two LEO constellations: LEO-600 and LEO-1200. This criterion corresponds to the minimum satellite count required to enable 3D positioning. The visibility analysis is conducted over a three-day simulation window, with a temporal resolution of 10 s. For each epoch, the number of satellites visible above a specified elevation mask is computed over a global latitude band ranging from -45° to $+45^\circ$ with 5° resolution. The probability is then calculated as the ratio of epochs with at least four visible satellites to the total number of epochs, averaged over all longitudes.

Figure 9 and Figure 10 illustrate the visibility probability for the LEO-600 and LEO-1200 constellations, respectively. For both systems, we observe that the probability decreases significantly with increasing elevation mask θ_0 , as expected due to the reduced satellite visibility cone. For low elevation masks (e.g., $\theta_0 = 10^\circ$), LEO-600 and LEO-1200 achieve near-complete visibility ($\geq 95\%$) across the entire latitude range. However, as θ_0 increases to 30° or more, the visibility probability drops sharply, particularly for the LEO-600 constellation.

At the equator (0° latitude), the probability of seeing at least four satellites above $\theta_0 = 10^\circ$ reaches approximately 0.98 for LEO-600 and nearly 1.00 for LEO-1200. This difference reflects the higher orbital altitude and larger footprint of the LEO-1200 satellites, which improves geometric diversity and coverage. The results highlight the advantage of using lower elevation masks in systems targeting global or equatorial coverage and the benefit of higher-altitude constellations in maintaining service availability under more stringent elevation constraints.

F. GDOP-BASED RESULTS

While no standardized terminology exists for defining navigation performance quality based on GDOP coefficient

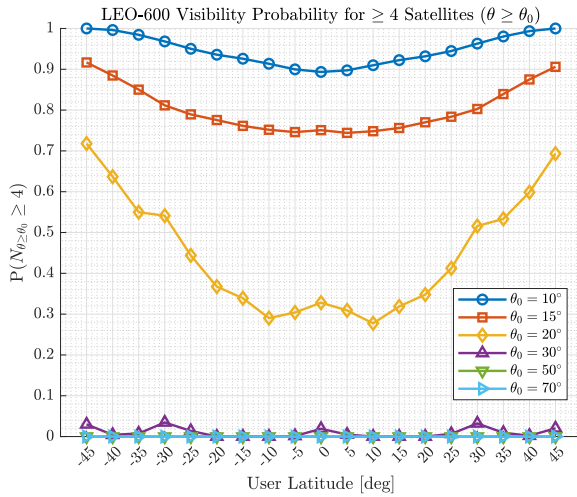


FIGURE 9. Visibility probability of observing at least four satellites from the LEO-600 constellation as a function of latitude and elevation mask θ_0 .

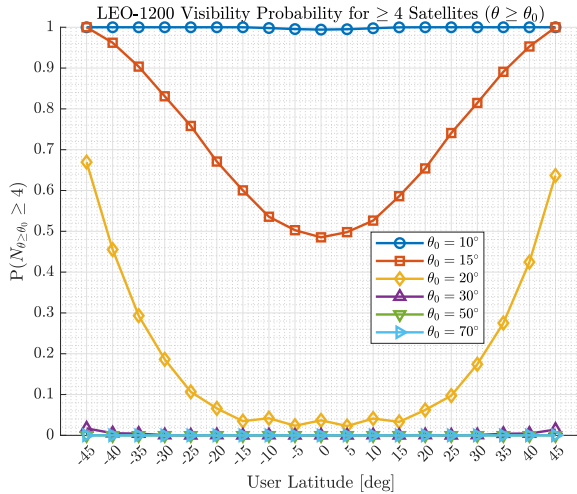


FIGURE 10. Visibility probability of observing at least four satellites from the LEO-1200 constellation as a function of latitude and elevation mask θ_0 .

TABLE 5. Interpretation of GDOP values.

GDOP	<1	1–2	2–5	5–10	10–20	>20
Rating	Ideal	Excellent	Good	Moderate	Fair	Poor

values, this study categorizes GDOP values in accordance with prior research [87], [88], as shown in Table 5.

The GDOP is computed for each MC trial for both LEO-600 and LEO-1200, with the results presented in Figure 11. In this analysis, outlier GDOP values are excluded to prevent statistical distortion. Since the GDOP formula involves the matrix inversion $(H^T H)^{-1}$, numerical instability can arise when H has nearly collinear rows, making the matrix nearly singular and leading to large numerical errors. This situation typically occurs when satellites are nearly aligned in a straight line.

TABLE 6. Outlier percentage by constellation.

Constellation	LEO-600	LEO-1200
Outlier percentage	0.53%	0.09%

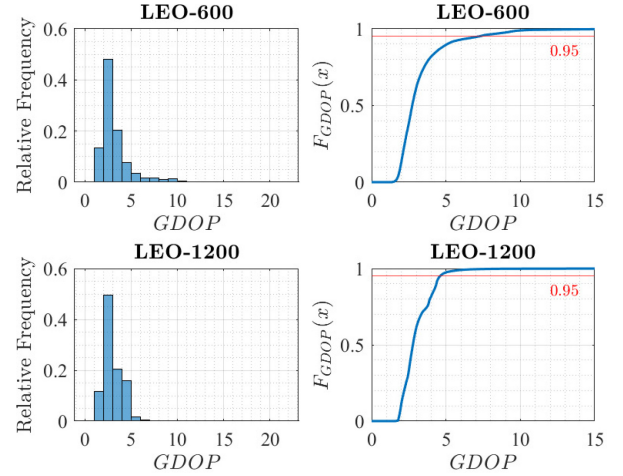


FIGURE 11. Histogram and Empirical Cumulative Distribution Function of GDOP for LEO-600 and LEO-1200 Constellations. A latitude mask of $[-45^\circ, +45^\circ]$ is applied.

A GDOP value is considered an outlier if it exceeds 20. The outlier ratios for both constellations are summarized in Table 6.

Both histograms, corresponding to the LEO-600 and LEO-1200 constellations, exhibit a right-skewed distribution, indicating that the majority of GDOP values are concentrated at lower levels, with a long tail extending toward higher values. However, the LEO-600 distribution features a more pronounced tail, with certain instances exceeding a GDOP value of 10. The peak of the distribution for both constellations occurs around an *Excellent* GDOP range (~ 1 – 2).

The Empirical Cumulative Distribution Function (ECDF) is computed for all GDOP values across both constellations, and the 95th percentile is determined, indicating the threshold below which 95% of the GDOP values fall. The 95th percentile threshold is approximately 7.1 (*Moderate*) for LEO-600 and 4.6 (*Good*) for LEO-1200. These results demonstrate that the LEO-1200 constellation exhibits a more reliable and stable GDOP performance, with fewer occurrences of high GDOP values. This can be attributed to the higher orbital altitude of LEO-1200 compared to LEO-600. Although LEO-1200 comprises fewer satellites, its greater geometric dispersion enhances positioning accuracy and reduces GDOP.

G. POSITIONING ACCURACY RESULTS

This section presents the evaluation of horizontal positioning accuracy based on the ECDF analysis for both PRS- and SSB-based configurations, considering LEO-600 and LEO-1200 constellations under FR1 and FR2 scenarios. The results are shown in Figure 12, and a zoomed-in view around the 95th percentile region is presented in Figure 13.

As shown in Figure 12, PRS-based configurations consistently outperform SSB-based configurations across all constellation and frequency scenarios. This can be attributed to the denser time-frequency occupancy of PRS signals, which enhances ToA estimation accuracy. In contrast, SSB-based positioning exhibits broader horizontal error distributions, particularly under LEO-600 and FR1 conditions.

Among all configurations, PRS with LEO-1200 under FR2 conditions achieves the best performance, with 95% of horizontal positioning errors below approximately 8 m. PRS with LEO-1200 under FR1 conditions follows closely, maintaining a sub-10 m accuracy. For LEO-600, PRS-based positioning results in higher errors, with 95th percentile values around 14 m (FR2) and 16 m (FR1).

Regarding SSB-based positioning, the LEO-1200 constellation again shows improved performance compared to LEO-600. The 95th percentile positioning errors are approximately 12 m (FR2) and 13 m (FR1) for LEO-1200, while they increase to 22 m (FR2) and 26 m (FR1) for LEO-600.

Figure 13 provides a zoomed-in view around the critical 95th percentile region, highlighting the sharp transitions of the ECDF curves. PRS-based curves exhibit steeper transitions compared to SSB-based curves, indicating a more consistent and tighter error distribution for PRS configurations.

These results confirm that:

- LEO-1200 constellations yield better positioning accuracy than LEO-600 for both PRS and SSB.
- FR2 scenarios outperform FR1 scenarios, mainly due to the larger available bandwidth, which improves ToA estimation precision.
- PRS signals significantly enhance positioning performance compared to SSB signals, reducing the 95th percentile error by up to 50% depending on the configuration.

In urban environments, GNSS positioning errors are often significantly degraded due to signal blockage and multipath effects, with horizontal errors commonly reaching tens of meters [89], [90]. Compared to this, the results obtained with PRS and SSB configurations in this study show substantially better performance. PRS-based positioning, particularly with LEO-1200 and FR2, achieves sub-10 m accuracy, outperforming typical urban GNSS results. Even SSB-based positioning, while exhibiting moderate degradation compared to PRS, maintains accuracy levels that are still superior to conventional urban GNSS performance.

In the context of 3GPP specifications for NTN, a positioning accuracy requirement of 10 km has been identified for network location verification [12]. All presented configurations, from the most accurate PRS LEO-1200 FR2 scenario to the higher-error SSB LEO-600 FR1 case, remain well within this 10 km requirement. Even the worst-case 95% error threshold of approximately 26 m is orders of magnitude below the 10 km limit.

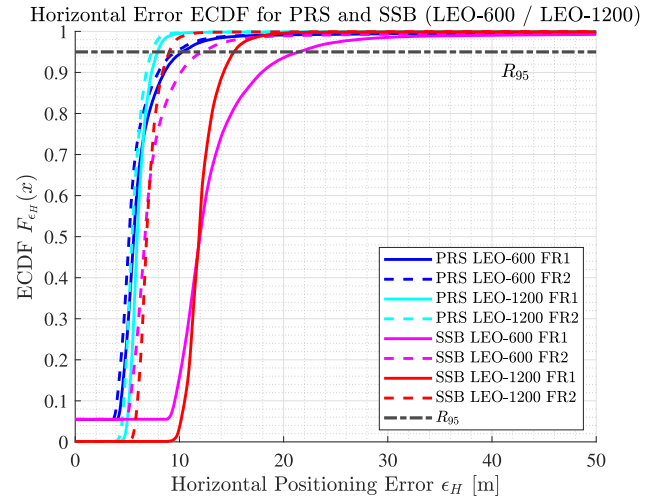


FIGURE 12. ECDF of horizontal positioning error for LEO-600 and LEO-1200 constellations using PRS and SSB, under FR1 and FR2 scenarios. The dashed red line marks the 95th percentile (R_{95}), indicating the error value below which 95% of the errors fall.

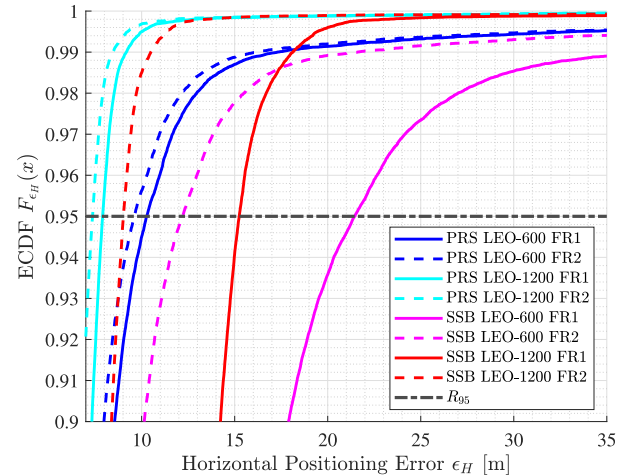


FIGURE 13. Zoomed-in view of the 95th percentile region in the ECDF of horizontal positioning error for PRS and SSB, considering LEO-600 and LEO-1200 constellations under FR1 and FR2 scenarios.

Further improvements in positioning accuracy can be achieved by adopting advanced error mitigation techniques. In particular, the use of dual-frequency receivers would allow for compensation of ionospheric propagation errors through direct estimation of the ionospheric delay. Additionally, multipath mitigation algorithms could reduce the impact of reflected signals on ToA estimation. These enhancements are expected to further improve 5G NTN positioning performance, potentially surpassing the accuracy levels typically achieved by terrestrial GNSS in challenging environments.

H. IMPLEMENTATION CONSIDERATIONS

While the analytical and simulation-based results presented in this work provide valuable insights into the achievable positioning performance of 5G NTN, it is equally important

to highlight several practical considerations that affect real-world deployment. In particular, three key aspects are discussed below: PRS resource overhead, hardware clock requirements, and acquisition and tracking under LEO dynamics.

PRS Resource Overhead: The accuracy of ToA-based positioning relies heavily on the density and bandwidth of the PRS configuration. However, these configurations directly impact spectral efficiency since PRS symbols occupy a fraction of the available time–frequency resources. Therefore, network operators must carefully balance the trade-off between enhanced positioning accuracy and overall system throughput when selecting PRS configurations.

Hardware Clock Requirements: In the context of receiver design, the choice of oscillator requires particular attention in LEO-based NTN systems. Unlike conventional GNSS, where signals are transmitted from MEO satellites with relatively slow Doppler dynamics, LEO satellites introduce significantly higher Doppler shifts and rates. These rapid variations impose additional stress on tracking loops, which can be further exacerbated by oscillator instability. Although the receiver clock bias is jointly estimated with the user position in ToA-based techniques, excessive short-term instability can degrade loop performance and ranging accuracy. For this reason, the use of more stable oscillators, such as OCXOs, is preferable over standard TCXOs to ensure robust positioning in LEO NTN scenarios.

Acquisition and Tracking Under LEO Dynamics: The high Doppler shifts and rates induced by LEO satellites tighten the design margins of acquisition and tracking loops compared to MEO-based systems. To maintain reliable ToA estimation, coherent integration times must be selected in conjunction with frequency-wipeoff accuracy and the expected Doppler rate (and jerk) to avoid loss of correlation. Practical implementations can benefit from aided acquisition (e.g., assisted frequency and time hypotheses), carrier-aided DLLs, and adaptive loop bandwidth strategies to enhance robustness under fast-varying channel conditions.

Overall, these considerations emphasize that, while PRS-based positioning in 5G NTN can achieve meter-level accuracy, its practical deployment will depend on carefully balancing resource overhead, oscillator stability, and robust signal tracking under LEO dynamics with the requirements of communication services.

V. CONCLUSION

This paper presented a comprehensive analysis of UERE and its impact on the positioning performance of 5G NTN utilizing both PRS and SSB. A complete UERE model was developed, considering satellite clock and ephemeris errors, receiver ToA measurement uncertainty, multipath propagation, and atmospheric effects across FR1 and FR2. In addition, the influence of satellite constellation geometry on positioning performance was assessed through GDOP and MC simulations for representative LEO constellations at

600 km and 1200 km. A minimum satellite visibility analysis was also carried out, confirming that both constellations can ensure sufficient coverage to guarantee at least four satellites in view for positioning feasibility across a wide range of elevation masks

Simulation results demonstrated that PRS-based positioning consistently outperforms SSB-based configurations across all scenarios, achieving 95th-percentile horizontal positioning errors as low as 8 m under favorable LEO-1200 FR2 conditions. Even SSB-based positioning, despite moderate accuracy degradation, achieved 95% error levels better than 26 m, comparable to conventional urban GNSS performance. All studied configurations satisfy the 3GPP NTN requirement of a maximum 10 km positioning error for network location verification.

The findings of this study provide valuable insights into the achievable positioning accuracy in 5G NTN systems and highlight the importance of signal design, link budget optimization, satellite visibility, and constellation configuration. Further improvements could be realized through advanced error mitigation techniques, such as dual-frequency receivers to correct ionospheric delays and multipath mitigation algorithms to reduce reflected signal impact on ToA estimation.

We also note that certain error components, such as ionospheric and tropospheric delays, are expected to diminish in 6G NTN systems if higher carrier frequencies are adopted, as currently envisioned in several research proposals. The same UERE framework and ToA-based positioning methodologies presented in this work could therefore be extended to evaluate the performance of future 6G NTN use cases, including autonomous aerial vehicles, high-precision industrial IoT, and global coverage positioning services.

Finally, future research could investigate the integration of Artificial Intelligence (AI)/Machine Learning (ML)-based techniques for dynamically estimating environment-dependent error components, such as multipath, NLOS, and atmospheric delays. Such hybrid model-based and data-driven frameworks could further enhance positioning robustness and accuracy in emerging 5G and upcoming 6G NTN systems, especially under adverse propagation and hardware conditions.

These insights support ongoing standardization efforts and pave the way for robust satellite-based PNT services in the 5G and beyond ecosystem.

APPENDIX A ACRONYMS

This paper uses an extensive number of acronyms, and to assist the reader, the following list presents all of them:

3G	Third Generation
3GPP	3rd Generation Partnership Project
4G	Fourth Generation
5G	Fifth Generation
6G	Sixth Generation
AI	Artificial Intelligence

AMF	Access and Mobility Function	HPA	High Power Amplifier
AoA	Angle of Arrival	IBO	Input Back-Off
AoD	Angle of Departure	ICAL	Integrated Communications and Localization
AR	Augmented Reality	ICI	Inter-Carrier Interference
ASIC	Application-Specific Integrated Circuit	IDFT	Inverse Discrete Fourier Transform
AWGN	Additive White Gaussian Noise	IFFT	Inverse Fast Fourier Transform
BER	Bit Error Rate	IIoT	Industrial Internet of Things
BFN	Beamforming Network	IMU	Inertial Measurement Unit
BO	Back-Off	IoT	Internet of Things
BW	Bandwidth	ISI	Intersymbol Interference
C/A	Civilian Acquisition	JCAP	Joint Communication And Positioning
CCDF	Complementary Cumulative Density Function	KPI	Key Performance Indicator
CMMB	China Mobile Multimedia Broadcasting	LCS	Location-based Services
CNR	Carrier-to-Noise Ratio	LEO	Low Earth Orbit
CP	Cyclic Prefix	LMC	Location Management Component
CRLB	Cramer-Rao Lower Bound	LMF	Location Management Function
CU	Centralised Unit	LNA	Low Noise Amplifier
DAC	Digital to Analog Converter	LOS	Line of Sight
DBF	Digital Beamforming	LPP	Localization Positioning Protocol
DFT	Discrete Fourier Transform	LPPa	Localization Positioning Protocol Annex
DL-AoD	Downlink Angle of Departure	LTE	Long Term Evolution
DL-OTDoA	Downlink Observed Time Difference of Arrival	MC	Monte Carlo
DLL	Delay Locked Loop	MCRB	Modified Cramer Rao Bound
DOP	Dilution of Precision	MEO	Medium Earth Orbit
DS	Delay Spread	ML	Machine Learning
DSSS	Direct-Sequence Spread Spectrum	MSB	Mean-Squared Bandwidth
DU	Distributed Unit	Multi-RTT	Multicell Round Trip Time
DVB	Digital Video Broadcasting	NF	Network Function
E-CID	Enhanced Cell ID	NGSO	Non Geostationary Satellite Orbit
ECEF	Earth-Centered, Earth-Fixed	NLOS	Non-Line of Sight
ECDF	Empirical Cumulative Distribution Function	NR	New Radio
EIRP	Effective Isotropic Radiated Power	NSGA-II	Non-dominated Sorting Genetic Algorithm II
EIRP	Equivalent Isotropic Radiated Power	NTN	Non-Terrestrial Network
E-SMLC	Evolved Serving Mobile Location Center	OCXO	Oven Controlled Crystal Oscillator
eNB	evolved NodeB	OFDM	Orthogonal Frequency-Division Multiplexing
EVM	Error Vector Magnitude	OTA	Over-The-Air
FLL	Frequency Locked Loop	OTDoA	Observed Time Differential of Arrival
FFT	Fast Fourier Transform	OTFS	Orthogonal Time Frequency Space
FFR	Full frequency re-use	PAPR	Peak-to-Average Power Ratio
FOV	Field of View	PLL	Phase-Locked Loop
FR	Frequency Region	PNT	Positioning, Navigation, and Timing
FR1	Frequency Range 1	POD	Precise Orbit Determination
FR2	Frequency Range 2	PPP	Precise Point Positioning
FR3	Frequency Range 3	PRN	Pseudo-Random Noise
FR4	Frequency Range 4	PRS	Positioning Reference Signal
FSPL	Free Space Path Loss	PSD	Power Spectral Density
GEO	Geostationary Orbit	PSS	Primary Synchronization Signal
GDOP	Geometrical Dilution of Precision	QoS	Quality of Service
GNSS	Global Navigation Satellite System	RAN	Radio Access Network
GNSSDO	GNSS Disciplined Oscillator	RAT	Radio-Access-Technology
GPS	Global Positioning System	RB	Resource Block
GPST	GPS Time	RE	Resource Element
gNB	Next Generation Base Station	RedCap	Reduced Capacity
GS	Ground Station	RG	Resource Grid
HAPS	High-Altitude Platform Systems	RMS	Root Mean Square
		RMSE	Root Mean Square Error

TABLE 7. Main notation and parameter definitions.

Symbol	Definition
σ_{Clk}	Std. dev. of satellite clock error
σ_{Eph}	Std. dev. of satellite ephemeris error
σ_{ToA}	Std. dev. of ToA measurement error
σ_{Multi}	Std. dev. of multipath error
σ_{Iono}	Std. dev. of residual ionospheric delay
σ_{Tropo}	Std. dev. of residual tropospheric delay
σ_{UERE}	Total User Equivalent Range Error
f_c	Carrier frequency
Δf	Subcarrier spacing
B	Signal bandwidth
B_n	DLL filter noise bandwidth
M	Number of OFDM symbols used
K	PRS comb size (subcarrier spacing multiple)
SNR_{inj}	Injected SNR in link budget
T_{Int}	Loop integration time
γ	Fraction of time PRS or SSB is present in T_{Int}
H	Satellites-receiver design matrix
Q	Covariance matrix of measurement errors
θ	Satellite elevation angle (degrees)
θ_0	Minimum elevation mask angle (degrees)
μ	5G numerology index
R_{95}	95 th percentile positioning error (horizontal)

RRC	Radio Resource Control
RTK	Real Time Kinematics
SBAS	Satellite Based Augmentation System
SIB	System Information Block
SIB19	System Information Block 19
SIC	Sequential Interference Cancellation
SFN	Single Frequency Network
SIR	Signal-to-Interference ratio
SINR	Signal-to-Interference plus Noise ratio
SLA	Service Level Agreement
SNR	Signal-to-Noise Ratio
SoO	Signal of Opportunity
SoP	Signal of Opportunity
SRS	Sounding Reference Signal
SRRC	Square Root Raised Cosine
SSB	Synchronization Signal Block
SSP	Subsatellite Point
SSS	Secondary Synchronization Signal
TA	Timing Advance
TC-OFDM	Time-Coded Orthogonal Frequency Division Multiplexing
TCXO	Temperature-Compensated Crystal Oscillator
TDL	Tapped Delay Line
TDoA	Time Difference of Arrival

TEC	Total Electron Content
TECU	TEC Units
TN	Terrestrial Network
ToA	Time of Arrival
ToF	Time of Flight
TR	Technical Report
TS	Technical Specification
AAV	Autonomous Aerial Vehicle
UE	User Equipment
UERE	User Equivalent Range Error
UL-AoA	Uplink Angle of Arrival
UL-TDoA	Uplink Time Difference of Arrival
UPA	Uniform Planar Array
USO	Ultra-Stable Oscillators
VR	Virtual Reality
VTEC	Vertical Total Electron Content
WDOP	Weighted Dilution of Precision
WLAN	Wireless Local Area Network
ZOH	Zero-Order Hold

APPENDIX B SYMBOLS DEFINITION

Table 7 summarizes the main symbols and parameters used throughout this article for clarity and consistency of notation.

REFERENCES

- [1] Mayank et al., "LEO-PNT feasibility aspects: Satellite navigation payload size, weight, and power analysis," *IEEE Access*, vol. 13, pp. 110069–110089, 2025, doi: [10.1109/ACCESS.2025.3583080](https://doi.org/10.1109/ACCESS.2025.3583080).
- [2] M. Hui et al., "A review of LEO-satellite communication payloads for integrated communication, navigation, and remote sensing: Opportunities, challenges, future directions," *IEEE Internet Things J.*, vol. 12, no. 12, pp. 18954–18992, Jun. 2025, doi: [10.1109/JIOT.2025.3553394](https://doi.org/10.1109/JIOT.2025.3553394).
- [3] W. Stock, R. T. Schwarz, C. A. Hofmann, and A. Knopp, "Survey on opportunistic PNT with signals from LEO communication satellites," *IEEE Commun. Surveys Tuts.*, vol. 27, no. 1, pp. 77–107, 1st Quart., 2025, doi: [10.1109/COMST.2024.3406990](https://doi.org/10.1109/COMST.2024.3406990).
- [4] K. Çelikbilek, E. Simona Lohan, and J. Praks, "Optimization of a LEO-PNT constellation: Design considerations and open challenges," *Int. J. Satell. Commun. Netw.*, vol. 43, no. 4, pp. 272–292, 2025, doi: [10.1002/sat.1555](https://doi.org/10.1002/sat.1555).
- [5] M. Shafi et al., "5G: A tutorial overview of standards, trials, challenges, deployment, and practice," *IEEE J. Sel. Areas Commun.*, vol. 35, no. 6, pp. 1201–1221, Jun. 2017, doi: [10.1109/JSAC.2017.2692307](https://doi.org/10.1109/JSAC.2017.2692307).
- [6] X. Lin, "The bridge toward 6G: 5G-advanced evolution in 3GPP release 19," *IEEE Commun. Stand.*, vol. 9, no. 1, pp. 28–35, Mar. 2025, doi: [10.1109/MCOMSTD.0001.2300063](https://doi.org/10.1109/MCOMSTD.0001.2300063).
- [7] M. El Jaafari, N. Chuberre, S. Anjuere, and L. Combelles, "Introduction to the 3GPP-defined NTN standard: A comprehensive view on the 3GPP work on NTN," *Int. J. Satell. Commun. Netw.*, vol. 41, no. 3, pp. 220–238, 2023, doi: [10.1002/sat.1471](https://doi.org/10.1002/sat.1471).
- [8] I. Rahman et al., "5G evolution toward 5G advanced: An overview of 3GPP releases 17 and 18," *Ericsson Technol. Rev.*, vol. 2021, no. 14, pp. 2–12, Oct. 2021, doi: [10.23919/ETR.2021.9904665](https://doi.org/10.23919/ETR.2021.9904665).
- [9] H. K. Dureppagari, C. Saha, H. S. Dhillon, and R. M. Buehrer, "NTN-based 6G localization: Vision, role of LEOs, and open problems," *IEEE Wireless Commun.*, vol. 30, no. 6, pp. 44–51, Dec. 2023.
- [10] "IMT towards 2030 and beyond (IMT-2030)." 2023. Accessed: Aug. 5, 2025. [Online]. Available: <https://www.itu.int:443/en/ITU-R/study-groups/rsg5/rwp5d/imt-2030/pages/default.aspx>
- [11] X. Lin, "An overview of 5G advanced evolution in 3GPP release 18," *IEEE Commun. Stand. Mag.*, vol. 6, no. 3, pp. 77–83, Sep. 2022, doi: [10.1109/MCOMSTD.0001.2200001](https://doi.org/10.1109/MCOMSTD.0001.2200001).

- [12] "Study on requirements and use cases for network verified UE location for non-terrestrial-networks (NTN) in NR," 3GPP, Sophia Antipolis, France, Rep. 38.882, 2022. [Online]. Available: https://www.3gpp.org/ftp/Specs/archive/38_series/38.882/38882-i00.zip
- [13] *Base Station (BS) Radio Transmission and Reception, Version 18.5.0 Release 18*, 3GPP Standard TS 38.104, 2024. [Online]. Available: https://www.etsi.org/deliver/etsi_ts/138100_138199/138104/16.04.00_60/ts_138104v160400p.pdf
- [14] S. Kang et al., "Cellular wireless networks in the upper mid-band," *IEEE Open J. Commun. Soc.*, vol. 5, pp. 2058–2075, 2024, doi: [10.1109/OJCOMS.2024.3373368](https://doi.org/10.1109/OJCOMS.2024.3373368).
- [15] N.-N. Dao et al., "A review on new technologies in 3GPP standards for 5G access and beyond," *Comput. Netw.*, vol. 245, May 2024, Art. no. 110370, doi: [10.1016/j.comnet.2024.110370](https://doi.org/10.1016/j.comnet.2024.110370).
- [16] *NG Radio Access Network (NG-RAN); Stage 2 Functional Specification of User Equipment (UE) Positioning in NG-RAN*, 3GPP Standard TS 38.305, Jun. 2022. Accessed: Aug. 27, 2025. [Online]. Available: https://www.etsi.org/deliver/etsi_ts/138300_138399/138305/17.00.00_60/ts_138305v170000p.pdf
- [17] A. Gonzalez-Garrido, J. Querol, H. Wymeersch, and S. Chatzinotas, "Interference analysis and modeling of positioning reference signals in 5G NTN," *IEEE Open J. Commun. Soc.*, vol. 5, pp. 7567–7581, 2024, doi: [10.1109/OJCOMS.2024.3503692](https://doi.org/10.1109/OJCOMS.2024.3503692).
- [18] R. M. Ferre and E. S. Lohan, "Comparison of MEO, LEO, and terrestrial IoT configurations in terms of GDOP and achievable positioning accuracies," *IEEE J. Radio Freq. Identif.*, vol. 5, no. 3, pp. 287–299, Sep. 2021, doi: [10.1109/JRFID.2021.3079475](https://doi.org/10.1109/JRFID.2021.3079475).
- [19] S. Emara, "Positioning in non-terrestrial networks," Ph.D. dissertation, Dept. Electr. Inf. Technol., Lund Univ., Lund, Sweden, 2021.
- [20] I. Lapin et al., "Code phase estimation with live 5g NR-NTN ka-band signal from LEO satellite using STARE," in *Proc. 11th Workshop Satell. Navig. Technol. (NAVITEC)*, Dec. 2024, pp. 1–5, doi: [10.1109/NAVITEC63575.2024.10843556](https://doi.org/10.1109/NAVITEC63575.2024.10843556).
- [21] M. Brambilla et al., "Integration of 5g and GNSS technologies for enhanced positioning: An experimental study," *IEEE Open J. Commun. Soc.*, vol. 5, pp. 7197–7215, 2024, doi: [10.1109/OJCOMS.2024.3487270](https://doi.org/10.1109/OJCOMS.2024.3487270).
- [22] B. Camajori Tedeschini et al., "A feasibility study of 5g positioning with current cellular network deployment," *Sci. Rep.*, vol. 13, no. 1, Art. no. 15281, 2023, doi: [10.1038/s41598-023-42426-1](https://doi.org/10.1038/s41598-023-42426-1).
- [23] K. Shamaei and Z. M. Kassas, "Receiver design and time of arrival estimation for opportunistic localization with 5G signals," *IEEE Trans. Wireless Commun.*, vol. 20, no. 7, pp. 4716–4731, Jul. 2021, doi: [10.1109/TWC.2021.3061985](https://doi.org/10.1109/TWC.2021.3061985).
- [24] F. Mooseli, S. Hayek, S. Kozhaya, and Z. M. Kassas, "Opportunistic navigation exploiting always-on and on-demand 5G downlink signals on a ground vehicle," in *Proc. IEEE 100th Veh. Technol. Conf.*, 2024, pp. 1–5, doi: [10.1109/VTC2024-Fall63153.2024.10757621](https://doi.org/10.1109/VTC2024-Fall63153.2024.10757621).
- [25] Z. Kassas and K. Shamaei, "Systems and methods for opportunistic time of arrival estimation for opportunistic localization with 5G signals," U.S. Patent 20230171142 A1, Jun. 2023.
- [26] A. Abdallah, J. Khalife, and Z. M. Kassas, "Exploiting on-demand 5G downlink signals for opportunistic navigation," *IEEE Signal Process. Lett.*, vol. 30, pp. 389–393, 2023, doi: [10.1109/LSP.2023.3234496](https://doi.org/10.1109/LSP.2023.3234496).
- [27] M. Neinavaie, J. Khalife, and Z. M. Kassas, "Cognitive opportunistic navigation in private networks with 5G signals and beyond," *IEEE J. Sel. Topics Signal Process.*, vol. 16, no. 1, pp. 129–143, Jan. 2022, doi: [10.1109/JSTSP.2021.3119929](https://doi.org/10.1109/JSTSP.2021.3119929).
- [28] E. D. Kaplan and C. Hegarty, *Understanding GPS/GNSS: Principles and Applications*, 3rd ed. Norwood, MA, USA: Artech House, 2017.
- [29] W. Lewandowski, J. Azoubib, and W. Klepczynski, "GPS: primary tool for time transfer," *Proc. IEEE*, vol. 87, no. 1, pp. 163–172, Jan. 1999, doi: [10.1109/5.736348](https://doi.org/10.1109/5.736348).
- [30] K. Wang and A. El-Mowafy, "LEO satellite clock analysis and prediction for positioning applications," *Geo-spatial Inf. Sci.*, vol. 25, no. 1, pp. 14–33, Jan. 2022, doi: [10.1080/10095020.2021.1917310](https://doi.org/10.1080/10095020.2021.1917310).
- [31] M. A. Lombardi, "The use of GPS disciplined oscillators as primary frequency standards for calibration and metrology laboratories," *NCSLI Measure*, vol. 3, no. 3, pp. 56–65, Sep. 2008, doi: [10.1080/19315775.2008.11721437](https://doi.org/10.1080/19315775.2008.11721437).
- [32] J. Bauer, C. Andrich, A. Ihlow, N. Beuster, and G. del Galdo, "Characterization of GPS disciplined oscillators using a laboratory GNSS simulation testbed," in *Proc. Joint Conf. IEEE Int. Freq. Control Symp. Int. Symp. Appl. Ferroelectr. (IFCS-ISAF)*, Jul. 2020, pp. 1–4, doi: [10.1109/IFCS-ISAF41089.2020.9234932](https://doi.org/10.1109/IFCS-ISAF41089.2020.9234932).
- [33] F. Kunzi and O. Montenbruck, "Precise onboard time synchronization for LEO satellites," *Navig., J. Inst. Navig.*, vol. 69, no. 3, Sep. 2022, Art. no. navi.531, doi: [10.33012/navi.531](https://doi.org/10.33012/navi.531).
- [34] (NovAtel Inc., Calgary, AB, Canada). *OEM719 Multi-Frequency GNSS Receiver*. Accessed: Feb. 26, 2025. [Online]. Available: <https://hexagondownloads.blob.core.windows.net/public/Novatel/assets/Documents/Papers/OEM719-Product-Sheet/OEM719-Product-Sheet.pdf>
- [35] F. S. Prol et al., "Position, navigation, and timing (PNT) through low earth orbit (LEO) satellites: A survey on current status, challenges, and opportunities," *IEEE Access*, vol. 10, pp. 83971–84002, 2022, doi: [10.1109/ACCESS.2022.3194050](https://doi.org/10.1109/ACCESS.2022.3194050).
- [36] K. Selvan, A. Siemuri, F. S. Prol, P. Välisuo, M. Z. H. Bhuiyan, and H. Kuusniemi, "Precise orbit determination of LEO satellites: A systematic review," *GPS Solut.*, vol. 27, no. 4, p. 178, 2023, doi: [10.1007/s10291-023-01520-7](https://doi.org/10.1007/s10291-023-01520-7).
- [37] L. Meng, J. Chen, J. Wang, and Y. Zhang, "Broadcast ephemerides for LEO augmentation satellites based on nonsingular elements," *GPS Solut.*, vol. 25, no. 4, p. 129, 2021, doi: [10.1007/s10291-021-01162-7](https://doi.org/10.1007/s10291-021-01162-7).
- [38] A. Hussain, A. Ahmed, M. A. Shah, S. Katyara, L. Staszewski, and H. Magsi, "On mitigating the effects of multipath on GNSS using environmental context detection," *Appl. Sci.*, vol. 12, no. 23, 2022, Art. no. 12389, doi: [10.3390/app122312389](https://doi.org/10.3390/app122312389).
- [39] E. Gill, J. Morton, P. Axelrad, D. M. Akos, M. Centrella, and S. Speretta, "Overview of space-capable global navigation satellite systems receivers: Heritage, status and the trend towards miniaturization," vol. 23, no. 17, p. 7648, 2023, doi: [10.3390/s23177648](https://doi.org/10.3390/s23177648).
- [40] "SpacePNT NaviLEO." satsearch. Accessed: Feb. 27, 2025. [Online]. Available: <https://satsearch.co/products/spacepnt-navileo>
- [41] *Radio Resource Control (RRC); Protocol Specification*, 3GPP Standard TS 38.331, 2025. [Online]. Available: <https://portal.3gpp.org/desktopmodules/Specifications/SpecificationDetails.aspx?specificationId=3213>
- [42] A. V. Oppenheim, R. W. Schaffer, and J. R. Buck, *Discrete-Time Signal Processing*, 2nd ed. Upper Saddle River, NJ, USA: Prentice Hall, 1999, p. 870.
- [43] J. J. Spilker Jr., P. Axelrad, B. W. Parkinson, and P. Enge, *Global Positioning System: Theory and Applications, Volume I*, P. Axelrad, B. W. Parkinson, P. Enge, and J. J. Spilker Jr., Eds., Reston, VA, USA: Amer. Inst. Aeronaut. Astronaut., 1996.
- [44] H. L. V. Trees, *Detection, Estimation, and Modulation Theory, Part I: Detection, Estimation, and Linear Modulation Theory*. Hoboken, NJ, USA: Wiley, Mar. 24, 2004.
- [45] S. M. Kay, *Fundamentals of Statistical Signal Processing*, vol. 1. Upper Saddle River, NJ, USA: Prentice Hall, 1993, p. 616.
- [46] J. A. del Peral-Rosado, J. A. López-Salcedo, G. Seco-Granados, F. Zanier, and M. Crisci, "Achievable localization accuracy of the positioning reference signal of 3GPP LTE," in *Proc. Int. Conf. Local. GNSS*, Jun. 2012, pp. 1–6, doi: [10.1109/ICL-GNSS.2012.6253127](https://doi.org/10.1109/ICL-GNSS.2012.6253127).
- [47] "NR; physical channels and modulation." Oct. 2020. Accessed: Feb. 21, 2025. [Online]. Available: <https://portal.3gpp.org/desktopmodules/Specifications/SpecificationDetails.aspx?specificationId=3213>
- [48] J. W. Betz and K. R. Kolodziejki, "Extended theory of early-late code tracking for a bandlimited GPS receiver," *Navigation*, vol. 47, no. 3, pp. 211–226, 2000, doi: [10.1002/j.2161-4296.2000.tb00215.x](https://doi.org/10.1002/j.2161-4296.2000.tb00215.x).
- [49] J. W. Betz and K. R. Kolodziejki, "Generalized theory of code tracking with an early-late discriminator part I: Lower bound and coherent processing," *IEEE Trans. Aerosp. Electron. Syst.* vol. 45, no. 4, pp. 1538–1556, Oct. 2009, doi: [10.1109/TAES.2009.5310316](https://doi.org/10.1109/TAES.2009.5310316).
- [50] F. Soualle et al., "New generation of PNT user terminals exploiting hybridization with LEO constellations," in *Proc. 37th Int. Techn. Meeting Satell. Div. Inst. Navig.*, 2024, pp. 887–947, doi: [10.33012/2024.19780](https://doi.org/10.33012/2024.19780).
- [51] F. Fabra, J. A. López-Salcedo, and G. Seco-Granados, "Analysis of baseband algorithms for LEO PNT," *Eng. Proc.*, vol. 54, no. 1, p. 36, 2023, doi: [10.3390/ENC2023-15458](https://doi.org/10.3390/ENC2023-15458).

- [52] H. Jiang and G. Gui, *Channel Modeling in 5G Wireless Communication Systems* (Wireless Networks). Cham, Switzerland: Springer, 2020.
- [53] "Study on channel model for frequencies from 0.5 to 100 GHz," 3GPP, Sophia Antipolis, France, Rep. 38.901, 2017. Accessed: Aug. 27, 2025. [Online]. Available: https://www.etsi.org/deliver/etsi_tr/138900_138999/138901/16.01.00_60/tr_138901v160100p.pdf
- [54] A. Zaidi, F. Athley, J. Medbo, U. Gustavsson, G. Durisi, and X. Chen, "Propagation & channel modeling," in *5G Physical Layer*, A. Zaidi, F. Athley, J. Medbo, U. Gustavsson, G. Durisi, and X. Chen, Eds., Cambridge, MA, USA: Academic, 2018, ch. 3, pp. 35–85, doi: [10.1016/B978-0-12-814578-4.00008-4](https://doi.org/10.1016/B978-0-12-814578-4.00008-4).
- [55] A. Alayón Glazunov, A. Razavi, R. Maaskant, and J. Yang, "Semi-analytical model of the Rician k-factor," *Radio Sci.*, vol. 55, no. 11, 2020, Art. no. e2020RS007099, doi: [10.1029/2020RS007099](https://doi.org/10.1029/2020RS007099).
- [56] Y. Lee and J. P. Choi, "Performance evaluation of high-frequency mobile satellite communications," *IEEE Access*, vol. 7, pp. 49077–49087, 2019, doi: [10.1109/ACCESS.2019.2909885](https://doi.org/10.1109/ACCESS.2019.2909885).
- [57] V. M. Baeza, E. Lagunas, H. Al-Hraishawi, and S. Chatzinotas, "An overview of channel models for NGSO satellites," in *Proc. IEEE 96th Veh. Technol. Conf.*, 2022, pp. 1–6, doi: [10.1109/VTCT2022-Fall57202.2022.10012693](https://doi.org/10.1109/VTCT2022-Fall57202.2022.10012693).
- [58] C. Amatetti et al., "Report on unified and data driven air-interface for 6G-NTN (1st version)," Eur. Union Horizon Eur., Brussels, Belgium, document 6G-NTN, D4.1, Jul. 2024. [Online]. Available: <https://www.6g-ntn.eu/download/d4-1-report-on-unified-and-data-driven-air-interface-for-6g-ntn/>
- [59] B. Hofmann-Wellenhof, H. Lichtenegger, and J. Collins, *Global Positioning System: Theory and Practice*. Vienna, Austria: Springer, Dec. 2012, p. 407.
- [60] L. Marini-Pereira, L. F. D. Lourenço, J. Sousasantos, A. O. Moraes, and S. Pullen, "Regional ionospheric delay mapping for low-latitude environments," *Radio Sci.*, vol. 55, no. 12, pp. 1–19, Dec. 2020, doi: [10.1029/2020RS007158](https://doi.org/10.1029/2020RS007158).
- [61] M. Imad, A. Grenier, X. Zhang, J. Nurmi, and E. S. Lohan, "Ionospheric error models for satellite-based navigation—Paving the road towards LEO-PNT solutions," vol. 13, no. 1, p. 4, 2024, doi: [10.3390/computers13010004](https://doi.org/10.3390/computers13010004).
- [62] M. Irsigler, G. W. Hein, and A. Schmitz-Peiffer, "Use of c-band frequencies for satellite navigation: benefits and drawbacks," *GPS Solut.*, vol. 8, no. 3, pp. 119–139, Jul. 2004, doi: [10.1007/s10291-004-0098-2](https://doi.org/10.1007/s10291-004-0098-2).
- [63] Q. Liu, C. Gao, Z. Peng, R. Zhang, and R. Shang, "Smartphone positioning and accuracy analysis based on real-time regional ionospheric correction model," *Sensors*, vol. 21, no. 11, p. 3879, 2021, doi: [10.3390/s21113879](https://doi.org/10.3390/s21113879).
- [64] R. Radovanovic, "Adjustment of satellite-based ranging observations for precise positioning and deformation monitoring," Ph.D. dissertation, Dept. Geom. Eng., Univ. Calgary, Calgary, AB, Canada, Jan. 2002. [Online]. Available: <http://hdl.handle.net/1880/42726>
- [65] B. Nava, P. Coisson, and S. M. Radicella, "A new version of the NeQuick ionosphere electron density model," *J. Atmos. Solar-Terrest. Phys.*, vol. 70, no. 15, pp. 1856–1862, 2008, doi: [10.1016/j.jastp.2008.01.015](https://doi.org/10.1016/j.jastp.2008.01.015).
- [66] Y. Yuan, N. Wang, Z. Li, and X. Huo, "The BeiDou global broadcast ionospheric delay correction model (BDGIM) and its preliminary performance evaluation results," *Navigation*, vol. 66, no. 1, pp. 55–69, 2019, doi: [10.1002/navi.292](https://doi.org/10.1002/navi.292).
- [67] M. R. Imad, "Atmospheric effects on satellite navigation systems: Ionosphere and troposphere," Ph.D. dissertation, Faculty Inf. Technol. Commun. Sci., Univ. Tampere, Tampere, Finland, 2024.
- [68] U. Ngayap et al., "Comparison of NeQuick g and klobuchar model performances at single-frequency user level," *Eng. Proc.*, vol. 54, no. 1, p. 7, 2023, doi: [10.3390/ENC2023-1547](https://doi.org/10.3390/ENC2023-1547).
- [69] M. Mäkelä, "Comparison and development of ionospheric correction methods in GNSS," M.S. thesis, Dept. Math., Tampere Univ. Technol., Tampere, Finland, 2016.
- [70] R. Orus, J. Parro, R. Prieto-Cerdeira, and R. Lucas (UNOOSA, Boulder, CO, USA), *NeQuickG Performance: 2 Year Global Results*. 2015. [Online]. Available: <https://www.unoosa.org/pdf/1cg/2015/1cg10/wg/wgb07.pdf>
- [71] T. Hobiger and N. Jakowski, "Atmospheric signal propagation," in *Springer Handbook of Global Navigation Satellite Systems*, P. J. Teunissen and O. Montenbruck, Eds., Cham, Switzerland: Springer Int. Publ., 2017, pp. 165–193, doi: [10.1007/978-3-319-42928-1_6](https://doi.org/10.1007/978-3-319-42928-1_6).
- [72] P. Misra and P. Enge, *Global Positioning System: Signals, Measurements, and Performance*. Lincoln, MA, USA: Ganga-Jamuna Press, 2011.
- [73] J. Saastamoinen, "Atmospheric correction for the troposphere and stratosphere in radio ranging satellites," in *The Use of Artificial Satellites for Geodesy*, Washington, DC, USA: Amer. Geophys. Union (AGU), 1972, pp. 247–251, doi: [10.1029/GM015p0247](https://doi.org/10.1029/GM015p0247).
- [74] H. S. Hopfield, "Tropospheric effect on electromagnetically measured range: Prediction from surface weather data," *Radio Sci.*, vol. 6, no. 3, pp. 357–367, Mar. 1971, doi: [10.1029/RS006i003p00357](https://doi.org/10.1029/RS006i003p00357).
- [75] J. A. Estefan and O. J. Sovers, "A comparative survey of current and proposed tropospheric refraction-delay models for DSN radio metric data calibration," Nat. Aeronaut. Space Admin., Washington, DC, USA, Rep. NASA-CR-197338, Oct. 1, 1994. (visited on 08/27/2025). [Online]. Available: <https://ntrs.nasa.gov/api/citations/19950013586/downloads/19950013586.pdf>
- [76] F. Dovis, B. Muhammad, E. Cianca, and K. Ali, "A run-time method based on observable data for the quality assessment of GNSS positioning solutions," *IEEE J. Sel. Areas Commun.*, vol. 33, no. 11, pp. 2357–2365, Nov. 2015, doi: [10.1109/JSAC.2015.2430513](https://doi.org/10.1109/JSAC.2015.2430513).
- [77] G. A. McGraw, "Tropospheric error modeling for high integrity airborne GNSS navigation," in *Proc. IEEE/ION Position, Locat. Navig. Symp.*, 2012, pp. 158–166, doi: [10.1109/PLANS.2012.6236877](https://doi.org/10.1109/PLANS.2012.6236877).
- [78] C.-S. Chen, Y.-J. Chiu, C.-T. Lee, and J.-M. Lin, "Calculation of weighted geometric dilution of precision," *J. Appl. Math.*, vol. 2013, no. 1, 2013, Art. no. 953048, doi: [10.1155/2013/953048](https://doi.org/10.1155/2013/953048).
- [79] H. Sairo, D. Akopian, and J. Takala, "Weighted dilution of precision as quality measure in satellite positioning," *IEE Proc. Radar, Sonar Navig.*, vol. 150, no. 6, p. 430, 2003, doi: [10.1049/ip-rsn:20031008](https://doi.org/10.1049/ip-rsn:20031008).
- [80] M. Kumm and O. Varis, "The world by latitudes: A global analysis of human population, development level and environment across the north-south axis over the past half century," *Appl. Geogr.*, vol. 31, no. 2, pp. 495–507, 2011, doi: [10.1016/j.apgeog.2010.10.009](https://doi.org/10.1016/j.apgeog.2010.10.009).
- [81] "Solutions for NR to support non-terrestrial networks (NTN)," Dec. 2019. Accessed: Aug. 27, 2025. [Online]. Available: <https://atisorg.s3.amazonaws.com/archive/3gpp-documents/Rel16/ATIS.3GPP.38.821.V1600.pdf>
- [82] A. Gavras et al., "Towards a space based infrastructure for 5G and beyond 5g networks," in *Proc. IEEE Globecom Workshops (GC Wkshps)*, 2022, pp. 1347–1352, doi: [10.1109/GCWkshps56602.2022.10008552](https://doi.org/10.1109/GCWkshps56602.2022.10008552).
- [83] "Discussion on performance evaluation for NTN," Huawei, Shenzhen, China, HiSilicon, Shenzhen, China, document TSG RAN WG1 Meeting #99, 3GPP R1-1911858, Nov. 2019. [Online]. Available: https://www.3gpp.org/ftp/TSG_RAN/WG1_RL1/TSGR1_99/Docs/R1-1911858.zip
- [84] I. del Portillo, B. G. Cameron, and E. F. Crawley, "A technical comparison of three low earth orbit satellite constellation systems to provide global broadband," *Acta Astronautica*, vol. 159, pp. 123–135, Jun. 2019, doi: [10.1016/j.actaastro.2019.03.040](https://doi.org/10.1016/j.actaastro.2019.03.040).
- [85] "Link budget results for NTN," Paris, France, Thales, document TSG RAN WG1 Meeting #99, 3GPP R1-1913351, 2019. [Online]. Available: https://www.3gpp.org/ftp/tsg_ran/WG1_RL1/TSGR1_99/Docs/R1-1913351.zip
- [86] *Physical Layer Procedures for Data, Version 18.2.0*, 3GPP Standard TS 2024. Accessed: Feb. 21, 2021. [Online]. Available: https://www.etsi.org/deliver/etsi_ts/138200_138299/138214/18.02.00_60/ts_138214v180200p.pdf
- [87] O. K. Isik, J. Hong, I. Petrunin, and A. Tsourdos, "Integrity analysis for GPS-based navigation of UAVs in urban environment," *Robotics*, vol. 9, no. 3, p. 66, 2020, doi: [10.3390/robotics9030066](https://doi.org/10.3390/robotics9030066).
- [88] M. Tahsin, S. Sultana, T. Reza, and M. Hossam-E-Haider, "Analysis of DOP and its preciseness in GNSS position estimation," in *Proc. Int. Conf. Electr. Eng. Inf. Commun. Technol. (ICEEICT)*, 2015, pp. 1–6, doi: [10.1109/ICEEICT.2015.7307445](https://doi.org/10.1109/ICEEICT.2015.7307445).

- [89] J. Lesouple, T. Robert, M. Sahmoudi, J.-Y. Tournet, and W. Vigneau, "Multipath mitigation for GNSS positioning in an urban environment using sparse estimation," *IEEE Trans. Intell. Transp. Syst.*, vol. 20, no. 4, pp. 1316–1328, Apr. 2019, doi: [10.1109/TITS.2018.2848461](https://doi.org/10.1109/TITS.2018.2848461).
- [90] Z. Zhang and L. Pan, "Current performance of open position service with almost fully deployed multi-GNSS constellations: GPS, GLONASS, galileo, BDS-2, and BDS-3," *Adv. Space Res.* vol. 69, no. 5, pp. 1994–2019, Mar. 2022, doi: [10.1016/j.asr.2021.12.002](https://doi.org/10.1016/j.asr.2021.12.002).



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